

University of Florence

School of Engineering - Department of Information Engineering

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Football Tracking and Player Identification from Broadcast Video

Candidate
Lorenzo Cappetti

Supervisors
Prof. Matteo Lapucci
Prof. Andrew David Bagdanov

Co-supervisors
Prof. Marco Sciandrone
Dott. Simone Magistri, PhD

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Abstract

Football analytics requires knowledge not only of player positions, but also of their identities. Unlike instrumented stadium tracking systems, broadcast video produces trajectories that are not directly linked to the match team sheet. This thesis addresses the problem of associating trajectories extracted from single-camera broadcast video with players identified by name.

The proposed system combines object detection and tracking, field calibration, team and role classification, and two jersey-number recognizers. The resulting evidence is integrated through a configurable decision layer, while the final identity is determined through a global assignment constrained by the team roster, with the option to abstain when the available information is insufficient.

The experiments show that errors cannot be attributed to a single bottleneck. External pre-training increases the overall jersey-number recognition accuracy from 32.2% to 57.1%, while changing the arbitration order between the two evidence sources improves end-to-end performance without retraining the models, with the effect also confirmed on the final test block. These results highlight the importance not only of the quality of individual components, but also of how their evidence is integrated.

Chapter 1

Introduction

Football analytics requires knowledge not only of each player's position, but also of their identity: distance covered, passing networks, and individual performance metrics are associated with a specific player rather than with an anonymous trajectory. Optical tracking systems installed in stadiums provide this information reliably, but they require dedicated infrastructure and are not available for a large proportion of recorded matches.

Broadcast video provides an alternative source of information, as it is available for every televised match and can be used without additional infrastructure. Extracting positional data from such footage is a widely studied problem supported by mature solutions, yet tracking from a single moving camera still introduces association errors and trajectory fragmentation. A further challenge is player identity: a tracking system (*tracker*) assigns arbitrary identifiers that are not directly linked to the players listed on the match team sheet. This thesis addresses this step by attempting to associate each trajectory with a specific player from the team roster, while allowing it to remain unidentified when the available evidence does not support a reliable decision.

1.1 Context and Motivation

Identifying a player in broadcast footage is not equivalent to a face recognition problem. Faces are rarely distinguishable at the scale at which players appear in the video, while the set of possible identities is known in advance from the match team sheets. The problem can therefore be formulated as a constrained assignment between trajectories and elements of a known set, rather than as an open-set recognition problem. The main difficulty lies in the visual evidence available to guide this assignment.

Among the available cues, team, role, and jersey number, the jersey number is the one most directly associated with identity, but it is also the most difficult to observe. Teammates wear visually similar uniforms, and the number is readable only in a limited subset of frames, when orientation, scale, and the absence of occlusions make it visible. Player identification therefore cannot be treated as an independent frame-by-frame classification problem: evidence must be accumulated along a trajectory so that a small number of informative observations can contribute to the identity of the entire tracklet.

This scarcity of evidence introduces a design choice that affects the whole system. When the available evidence is insufficient, the system can either force a decision or abstain. These two outcomes do not have the same cost: an incorrect identity can distort statistics associated with multiple players, whereas a missing identity primarily results in a loss of coverage. For this reason, the system explicitly supports abstention from identity assignment. The experiments presented later show, however, that the criteria used to accept and prioritize evidence can themselves become a source of error.

1.2 Problem Definition

The objective is to transform a broadcast recording, together with the rosters of the two teams, into a sequence of observations in which each visible player is localized and, whenever possible, identified by name. The problem is addressed through three main stages:

- **Detection and tracking:** detect the players visible in each frame and associate observations over time to construct coherent trajectories;
- **Evidence extraction:** associate each trajectory with the information useful for identification, including field position, team, role, and jersey number;
- **Identity assignment:** integrate the available evidence and associate each trajectory with a player from the roster, while retaining the option to abstain when the available information does not support a reliable decision.

This decomposition is also useful for error analysis, since the stages are not independent: a fragmented trajectory, or one that combines observations from different players, can compromise the subsequent identification stages.

1.3 Contributions

This work develops and evaluates a complete pipeline for player identification from broadcast video. The main contributions of the thesis are:

- **End-to-end pipeline:** integration of object detection, tracking, camera calibration, team and role classification, jersey-number recognition, and identity assignment.

- **Separation between evidence and decision:** the information produced by the different components is kept separate and combined through an explicit and configurable decision layer, which also supports abstention when the available evidence is insufficient.
- **Arbitration between recognizers:** two complementary jersey-number recognition sources are compared under different priority policies. The experiments show that changing the arbitration order can improve end-to-end performance without modifying or retraining the recognizers.
- **Experimental analysis of the pipeline:** controlled comparisons, oracle substitutions, and out-of-domain evaluations are used to identify the main sources of error and to distinguish effective improvements from interventions that do not yield measurable benefits.

The remainder of the thesis is organized as follows. Chapter 2 reviews the relevant literature and datasets; Chapter 3 describes the proposed architecture; Chapter 4 defines the experimental protocol and evaluation metrics; Chapter 5 presents the experimental results; and finally, Chapter 6 discusses their implications, limitations, and possible directions for future work.

Chapter 2

Related Work and Background

Reconstructing who is present on the field can be decomposed into three problems addressed in sequence: detecting athletes in each frame, tracking each of them over time, and determining which person each trajectory belongs to [22]. Since each stage relies on the output of the previous one, upstream errors can strongly limit what subsequent stages are able to recover. An athlete who is never detected cannot be identified downstream, while a trajectory that merges observations from different players can compromise even an accurate identification model.

In the geometric formulation of association adopted by many trackers, the quality of detector localization directly affects temporal association as well, making the choice of detector relevant to both detection and tracking [27]. One possible strategy for handling interrupted trajectories is appearance-based re-identification, but this encounters a structural limitation in team sports, where identical uniforms make appearance informative for distinguishing teams but less discriminative for distinguishing teammates [18].

A cue more directly associated with identity is the jersey number, which is usually readable from the back, over a small portion of an already small

crop, and only in a handful of frames out of several hundred. Reading it first requires locating it, and the different approaches used to obtain this region trade annotation cost against dependence on the specific footage on which they were developed. When annotated data are scarce, treating the jersey number as a sequence of digits rather than as one of roughly one hundred classes is a useful formulation [6, 20]. Field calibration then provides metric positions and, with them, the possibility of constraining association according to physical plausibility rather than pixel proximity [24].

Two consequences follow for the remainder of the thesis. First, the benchmark that defines the task also provides the reference framework against which this work is evaluated, and its annotation conventions determine how the results in Chapter 5 should be interpreted; this chapter therefore begins with the datasets. Second, because the pipeline studied here is largely assembled from components available in the literature, the boundary between reviewed work and the contribution of this thesis should remain explicit rather than implicit. Each section therefore concludes with a paragraph entitled *In this work*, specifying which of the reviewed methods are actually used in the system and in which configuration. Statements regarding how a component is implemented in this work, as opposed to how it is described in the literature, are confined to those paragraphs and to Chapter 3.

2.1 Datasets

SoccerNet Game State Reconstruction

SoccerNet-GSR [22] is the main dataset used in this thesis. It extends SoccerNet-Tracking, which provides 200 thirty-second broadcast clips annotated with bounding boxes for players, referees, and the ball, by adding the

annotations required for the game state reconstruction task: more than 9.37 million field-line points for pitch localization and camera calibration, and more than 2.36 million athlete positions on the field, each associated with role, team membership, and jersey number. The clips are divided into training, validation, and test sets, together with an additional separate challenge set. Since the sequences consist of uncut broadcast footage captured by a single moving camera, only part of the field is visible at any given time, and the task is consequently limited to athletes within the camera's field of view.

The dataset adopts several annotation conventions that are relevant to the interpretation of the results:

- **Roles and teams.** Athletes are classified as *player*, *goalkeeper*, *referee*, or *other*; team membership is annotated only for players and goalkeepers as *left* or *right*, according to the position of the goals relative to the camera.

- **Jersey number per tracklet.** If a number is readable in at least one frame, it is associated with the entire tracklet; otherwise, it is annotated as *null*. Its presence in an annotation therefore does not imply that it is visible in that specific frame.

- **Athlete identification.** Role, team, and jersey number are not necessarily unique; for this reason, the dataset also provides a conventional *track identifier* used to associate predictions with annotated athletes.

Jersey Number Dataset, SJN-210K

For jersey-number recognition, this work also uses the SJN-210K dataset by Li et al. [13], which is employed for the external pre-training evaluated in Section 5.4.1. The dataset contains approximately 208,000 player crops, of

which 21,410 contain a visible number (14,956 in the training set and 6,454 in the test set); 12,746 of these also include a quadrilateral annotation of the jersey-number region.

The main characteristic of SJN-210K is its data diversity. Since it is not restricted to a single broadcast production style, it includes different competitions, uniforms, typefaces, and recording conditions. This diversity makes it complementary to SoccerNet-GSR, whose data instead come from a more homogeneous broadcast distribution. Figure 2.1 shows a comparison between the two sources.



(a) SoccerNet-GSR dataset (b) SJN-210K [13]

Figure 2.1: Examples of annotated jersey-number regions in SoccerNet-GSR and SJN-210K.

Two subsets are derived from the original dataset while preserving its training/test partition. The first, used for jersey-number region detection, contains 12,727 usable quadrilateral-annotated crops after preprocessing: 8,938 images from the original training partition and 3,789 images from the original test partition, the latter used here as validation data. The second, used for recognition, contains all 21,410 positive crops together with their corresponding transcriptions.

The recognition subset nevertheless exhibits a strongly imbalanced class

distribution: only 63 distinct jersey numbers are present in the training set and 59 in the test set, with some classes occurring much more frequently than others. This makes the dataset particularly suitable for pre-training models based on individual digits, while making it less appropriate for a classifier that must directly distinguish among all possible jersey numbers. The 186,685 negative crops in the original dataset, which do not contain a readable number, are not used in this work.

A further reference is the ice-hockey dataset introduced by Koshkina and Elder [11], which demonstrates the possibility of transferring jersey-number recognition across different sports. The related results are discussed in Section 2.5.

Custom Broadcast Footage

Public sports datasets are homogeneous by construction: SoccerNet-GSR is derived from a common broadcast production style, with similar camera placement, framing conventions, and image processing across clips. Such homogeneity is desirable in a benchmark because it isolates the task from nuisance variation, but it also means that performance measured on the benchmark alone does not characterize performance on generic football broadcast footage.

To investigate this aspect, a small collection of broadcast recordings outside SoccerNet was gathered, including domestic and international matches with different uniform styles, number typefaces, lighting conditions, and camera framing. These recordings do not provide positional *ground truth* and therefore cannot be used to evaluate tracking. Instead, two forms of annotation were produced: a team roster for each match and Identity Benchmark V1, a set of manually verified identity labels annotated blindly with respect

to model predictions and maintained under version control. Section 5.5.3 uses these labels to evaluate the roster constraint, while Section 5.6 uses the recordings to measure how far the region detector transfers beyond the SoccerNet domain.

In this work. SoccerNet-GSR [22] provides all *ground truth* used for tracking, calibration, and jersey-number evaluation. Its training set provides the data used both to train the components and to construct the held-out evaluation blocks; its test set is used only twice, for the recognizer comparison in Section 5.4.1 and for the final end-to-end evaluation in Section 5.7.4, while remaining separate from development decisions: the end-to-end block is not used to select the policy, modify thresholds, or choose checkpoints.

The jersey-number dataset by Li et al. [13] is used exclusively for pretraining, in two derived forms: a jersey-region detection set containing 8,938 training images and 3,789 validation images, and a recognition set of cropped jersey numbers with their corresponding transcriptions. The custom recordings and Identity Benchmark V1 are used for the roster and domain-transfer experiments. The exact subsets used in each experiment, and the order in which they are used, are methodological rather than descriptive details and are documented in Section 4.1.

2.2 Object Detection

Object detection is the first step in localizing the elements present in each frame and provides the data required by the subsequent tracking and identification stages. One-stage detectors, introduced by YOLO [19], directly perform classification and bounding-box regression in a single forward pass, with an emphasis on inference speed. This property makes them particularly

suitable for video, where detection must be repeated for every frame. The SoccerNet-GSR baseline also uses a YOLO detector [22].

In tracking, however, performance depends not only on the number of detected objects, but also on the accuracy of their coordinates. Localization errors can propagate to the association stage between consecutive frames. Detector selection therefore involves a trade-off between accuracy, recall, and computational cost, whose impact on the system is analyzed in Section 5.1.

The implementation relies on the Ultralytics model family [10], which provides a common interface for different computer vision tasks. This choice makes it possible to use the same infrastructure for several models within the pipeline.

In this work. The Ultralytics family [10] is used for four distinct components:

- **Person detector:** fine-tuned on SoccerNet-GSR, it localizes athletes and provides detections to the tracker.
- **Jersey-number region detector:** it localizes the jersey number within the athlete crop. The region detector introduces a dependency on the distribution of the annotated data; its ability to transfer to different broadcast productions is analyzed in Sections 5.4.2 and 5.6.
- **Crop classifiers:** two models trained on manually annotated crops are used to analyze frame selection in Section 5.4.4.
- **Pose estimator:** used off the shelf, it provides an independent signal about athlete orientation in the same analysis.

The different generalization capabilities of these components are discussed in Chapter 6.

2.3 Multi-Object Tracking

Modern multi-object tracking largely follows the tracking-by-detection paradigm: a detector localizes objects in each frame, and an association module links detections over time to form trajectories [27]. Tracking quality therefore depends both on the availability of detections and on the correctness of their temporal association.

ByteTrack [27] is the tracker adopted in this work. The method uses a two-stage association strategy. High-confidence detections are associated first, after which low-confidence detections are used to recover tracklets that remain unmatched. This avoids discarding potentially useful observations in the presence of occlusions. The association is mainly geometric: a Kalman filter predicts tracklet positions, and matching is based on Intersection-over-Union (IoU), without using appearance embeddings.

Association errors can produce two main consequences:

- **Fragmentation:** a trajectory is interrupted and the same athlete is represented by multiple tracklets, splitting the evidence available for subsequent identification.
- **Identity switch:** a tracklet becomes associated with the wrong person, combining observations from different players within a single trajectory.

An alternative approach integrates visual appearance directly into the association process. StrongSORT [3], which is used by the SoccerNet-GSR baseline [22], combines tracking with a person re-identification model. In the system considered in this thesis, these two aspects are instead kept separate: ByteTrack constructs tracklets using geometric information, while appearance information is subsequently used by the tracklet-linking module

described in Section 3.2.3. This separation makes it possible to analyze the contribution of tracking and linking independently.

In this work. Tracking is performed by ByteTrack [27] on the detections produced by the detector described in Section 3.1, followed by the tracklet-linking stage described in Section 3.2.3. Tracking performance, including identity switches and fragmentation, is evaluated in Section 5.1.

2.4 Person Re-Identification and Tracklet Association

Person Re-Identification (ReID) represents individuals through embeddings in which crops of the same person are close to one another, allowing observations from different frames or tracklets to be associated. In football, however, visual appearance is strongly shared among teammates: uniform color, texture, and design are informative for distinguishing teams, but provide limited information for distinguishing individual players. ReID is therefore a useful signal for tracklet association, but it is not sufficient on its own for individual identification. The jersey number, discussed in Section 2.5, provides a complementary and more identity-specific cue.

The literature mainly addresses two aspects of this problem:

- **Robustness to occlusion.** BPBreID [21] uses part-based representations, associating features and visibility information with different body parts. PRTReID [18] extends this approach to the sports domain by jointly learning identity, team, and role. The model is also used by the SoccerNet-GSR baseline [22].

- **Post-processing correction of tracklets.** GTA [23] introduces a Splitter, which separates multiple identities contained within the same tracklet by clustering embeddings with DBSCAN [4], and a Connector, which links distinct tracklets belonging to the same athlete. Using both modules increases HOTA from 79.41% to 83.11% on SoccerNet [23].

These approaches show that appearance information can be used not only during tracking, but also to correct or consolidate trajectories after they have been generated.

In this work. The pipeline uses handcrafted HSV color histograms as an appearance signal for team clustering and as a gate in the tracklet linker. PRTReID [18] is integrated as an optional source of embeddings for auxiliary analyses, but it is not the main linker evaluated in Chapter 5. GTA is instead retained as a reference from the literature: neither its Splitter nor a ReID-based Connector is part of the final configuration.

The linking strategy evaluated experimentally uses the appearance and geometric signals described in Chapter 3. The results in Section 5.2 show that, under the conditions considered, more accurate geometry alone is not sufficient to improve association.

2.5 Jersey Number Recognition

The jersey number is the main source of information for distinguishing players within the same team, but it is difficult to read because it occupies a small region of the crop, may be occluded or blurred, and is mainly visible when the athlete is facing away from the camera. Its absence in a single frame therefore does not imply that the number is unreadable elsewhere in

the tracklet.

The main methodological choices concern three stages:

- **Recognition formulation.** The jersey number can be treated as a class belonging to a predefined set [5,26], or as a sequence of digits [20]. The latter formulation makes it possible to learn individual digits and to recognize combinations that were not observed during training.
- **Prediction decoding.** Sequential approaches include CTC models [6, 20] and autoregressive recognizers based on attention [14]. A CTC model can assign a score to different candidate strings without repeating inference, which is useful when prior information is subsequently used to rank possible readings.
- **Evidence localization and selection.** The region containing the jersey number can be obtained through learned localization [13], dedicated detectors, or geometric information derived from pose [11, 16]. Alternatively, keyframe or readable-crop selection can restrict recognition to the most informative observations [1, 11].

Predictions obtained from individual frames are generally aggregated along the tracklet, for example through voting [22]. Tracking quality therefore also affects recognition: fragmentation reduces the available evidence, while incorrect associations can combine observations belonging to different players.

In this work. The pipeline uses two complementary recognition paths:

- **Generic OCR**: MMOCR [12], with DBNet and SAR [14, 15], and EasyOCR [9] as an alternative;

- **Region + CTC**: a dedicated detector localizes the jersey number, which is then recognized by a CRNN-CTC [6, 20] based on ResNet-18 [7] and a bidirectional LSTM [8].

The region detector introduces a dependency on annotated data, which is analyzed in Section 5.6. The CRNN-CTC recognizer is instead pre-trained on the dataset by Li et al. [13], as evaluated in Section 5.4.1. The predictions produced by the two paths are kept as separate pieces of evidence and combined at tracklet level through an explicit decision policy.

2.6 Field Calibration

Tracking in image space produces trajectories expressed in pixels, whereas distances, speeds, and tactical positions are defined with respect to the playing field. Since the field is planar, the mapping between the two coordinate systems can be described by a homography, estimated from visible field features and their correspondences in the field model [24].

In broadcast footage, calibration is made difficult by camera motion, partial field visibility, and the limited number of distinctive geometric elements. TVCalib [24], which is also used by the SoccerNet-GSR baseline [22], addresses the problem by segmenting field lines and estimating camera parameters through reprojection error.

The metric coordinates obtained from calibration can be used for two main purposes:

- **Field localization**: transform athlete observations into metric coordinates, which are required for game state reconstruction [25];
- **Tracklet association**: assess the physical plausibility of a possible

merge by penalizing associations that would require movements incompatible with athlete motion.

The ablation study of the SoccerNet-GSR baseline also shows that perfect calibration alone does not eliminate the main sources of error in the pipeline [22]. The role of calibration and the effectiveness of using it for linking are analyzed in Sections 5.3 and 5.2.

In this work. The pipeline supports two calibration sources:

- **TVCalib** [24], used as the primary method;
- **field quadrilateral**, used as a fallback when the primary calibration is unavailable.

The final configuration uses TVCalib for metric evaluations and GS-HOTA. The resulting coordinates are also provided to the *tracklet linker*, while jersey number recognition remains independent of calibration and operates in image space.

2.7 Positioning of This Work

The system described in this thesis builds upon the SoccerNet-GSR baseline [22], from which it inherits a large proportion of its components. This overlap is intentional: the objective is not to propose a new architecture, but to analyze the decisions made by a pipeline of this kind and quantify their impact. Table 2.1 summarizes the main differences.

The most relevant difference concerns arbitration. The baseline uses a single recognition path, whereas this pipeline keeps the evidence produced by two sources with different error profiles separate from the final decision. The

Stage	GSR Baseline [22]	This work
Detection	YOLOv8 [10]	YOLO26 [10], with fine-tuning
Tracking	StrongSORT [3]	ByteTrack [27]
Tracklet linking	none	appearance + geometry gates
Visual appearance	PRTReID [18]	HSV histograms, optional PRTReID
Calibration	TVCalib [24]	TVCalib as primary method, field quadrilateral as fallback
Jersey number	MMOCR [12]	MMOCR and region-based CRNN-CTC
Aggregation	majority voting	evidence and candidates
Identity	attributes	attributes + roster constraint/global assignment
Arbitration	implicit	explicit decision layer

Table 2.1: Comparison between the reference baseline and the proposed system. The main differences concern tracking, tracklet linking, jersey number recognition, and decision mechanisms; these are the focus of the experiments presented in Chapter 5.

choice of which source should take priority in case of disagreement therefore becomes an explicit and experimentally controllable decision, analyzed in Section 5.5. The same applies to the roster constraint, whose effectiveness is evaluated against manually annotated identities in Section 5.5.3.

Since a large proportion of the components are inherited from the literature, the attribution of results is also a relevant methodological issue: an improvement may depend either on the introduced modification or on the evaluation protocol. Chapter 4 therefore describes the measures adopted to isolate these effects, which are also necessary for interpreting the negative results reported in Chapter 5.

The end-to-end results, expressed in GS-HOTA according to the SoccerNet-GSR protocol, are computed on twelve sequences rather than on the complete benchmark partitions. The comparison with the published baseline therefore serves only as contextual reference, since the evaluated subset, metric implementation, and component configuration differ. Causal attribution of the improvement instead relies on the paired comparison between arm A and arm B, in which only the priority of the evidence sources is changed. The

contribution of this work lies in analyzing and quantifying the constraints of an inherited identification pipeline and, in particular, in showing that one of the largest improvements obtained results from the decision strategy rather than from additional model training.

Chapter 3

System Architecture

This chapter describes the architecture of the proposed pipeline, which progressively transforms video frames into enriched tracklets and, when possible, into identified players. The process is organized into four main blocks:

- **Perception and tracking:** player detection and construction of anonymous tracklets;
- **Tracklet enrichment:** estimation of field position, team, and role, together with trajectory consolidation;
- **Recognition and evidence:** production and arbitration of jersey-number evidence;
- **Identity assignment:** combination of the available evidence with roster constraints to determine the player's identity.

The architecture keeps three levels separate: *observation*, *decision*, and *identity*. Recognition modules produce evidence without directly assigning a player; a configurable policy selects the operational evidence; finally, the

global assignment combines this evidence with contextual information and the roster.

Figure 3.1 summarizes the pipeline flow. The following sections describe its components in processing order.

3.1 Perception and Tracklet Construction

The first stage of the pipeline transforms the video sequence into a representation of anonymous trajectories. Each frame is processed by a detector, and the resulting detections are subsequently associated over time by a tracker. The output is a set of *tracklets*, namely temporally coherent sequences of observations associated with a tracker identifier but not yet linked to a player identity.

This representation constitutes the processing unit for the subsequent stages. The front-end therefore prioritizes recall: a missed detection can interrupt a trajectory and prevent downstream modules from recovering the lost information, whereas a spurious detection can still be filtered using the information and constraints available later in the pipeline.

3.1.1 Object Detection

Detection is performed using a one-stage detector from the YOLO family, fine-tuned on SoccerNet-GSR. The class space is deliberately restricted to *person* and *ball*. The distinction between outfield players, goalkeepers, and referees is not made at this stage, but is deferred to the appearance-based classification described in Section 3.2.2.

This choice avoids introducing a semantic distinction at a stage whose main objective is to preserve trajectories. A classification error in the detector

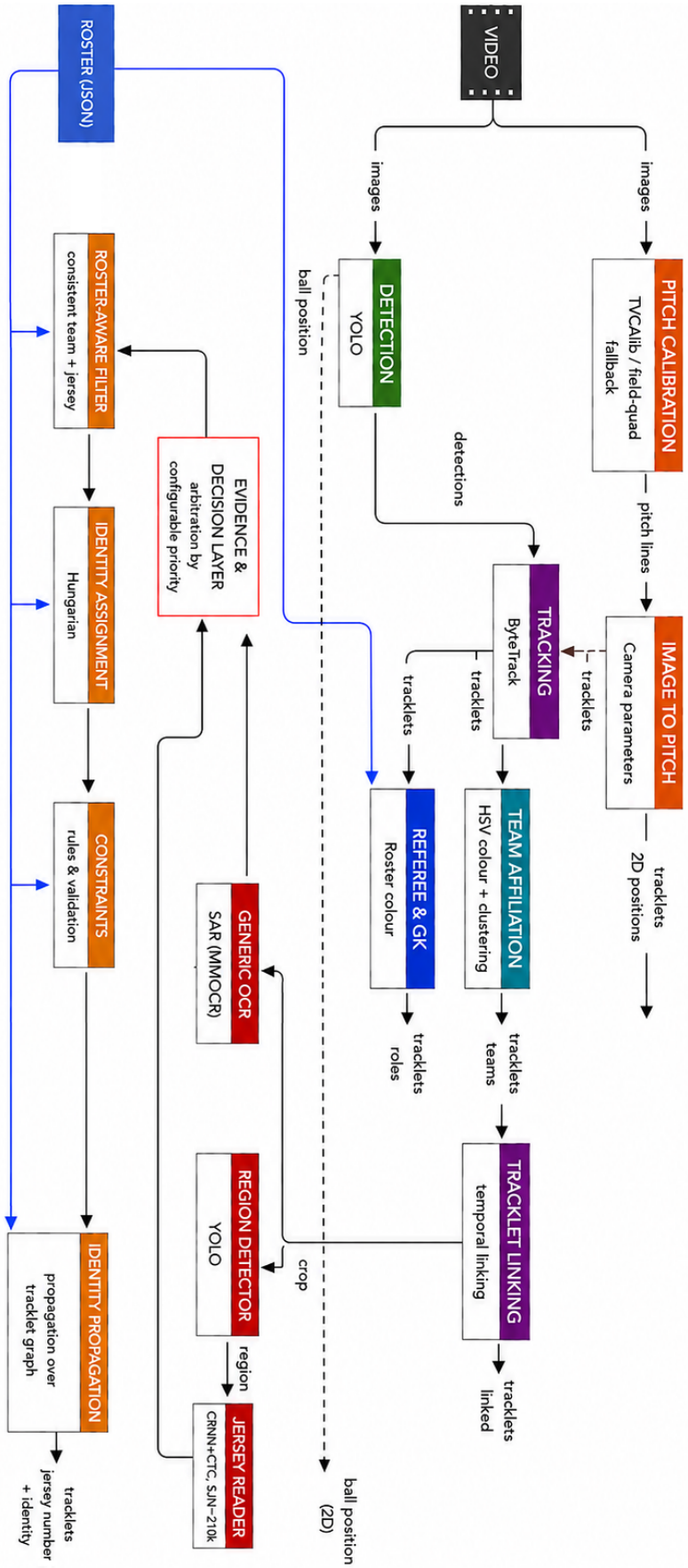


Figure 3.1: Overall architecture of the proposed pipeline.

could propagate to the tracker and fragment a trajectory, whereas the same distinction can be reassessed later using appearance information aggregated over the tracklet.

The detector operates with a confidence threshold of 0.05 for the *person* class, chosen to prioritize recall. The ball is detected using an even lower threshold of 0.002 and is also subject to a bounding-box area ratio constraint, which helps reject detections that are incompatible with its expected relative size.

3.1.2 Temporal Association

Detections are associated over time using **ByteTrack**. The tracker adopts a two-stage strategy: high-confidence detections are associated first, after which low-confidence detections are used to recover associations with tracklets that remain unmatched. This strategy is well suited to football footage, where occlusions and variations in visual quality can temporarily reduce detection confidence without corresponding to the player actually leaving the scene.

The configuration uses a track activation threshold of 0.10, requires at least two consecutive frames before emitting a tracklet, and temporarily keeps lost tracklets active for up to 150 frames. The latter parameter helps preserve a trajectory during short absences from the camera's field of view.

The identifier produced by the tracker, `raw_track_id`, therefore represents a temporary trajectory identity rather than the player's identity. The resulting tracklets provide the input to the enrichment and consolidation stages described in the following sections. In particular, *tracklet linking* can subsequently merge multiple temporary identifiers into the same consolidated trajectory, represented by `display_track_id`.

3.2 Tracklet Localization and Enrichment

The tracklets produced by the front-end describe trajectories in image space, but do not yet contain sufficient information about field position, team, role, or continuity beyond tracker interruptions. This stage enriches them through three components:

- **field calibration**, to obtain metric coordinates;
- **team and role classification**, to add semantic information;
- **tracklet linking**, to consolidate compatible fragments of the same trajectory.

3.2.1 Field Localization and Calibration

Detection and tracking operate in image space, where coordinates depend on camera perspective. To obtain a metric representation, the pipeline estimates a transformation between the image plane and the field plane.

The primary calibration method is TVCalib, which uses visible field markings and the known geometry of the playing field to estimate camera parameters. Since broadcast footage includes pans and zooms, calibration is updated over time, and each observation is associated with the temporally closest available estimate.

The position of each athlete on the field is obtained by projecting the midpoint of the lower edge of the bounding box, which is used as an approximation of the contact point with the ground. When TVCalib is not available, the pipeline can use a fallback estimate based on the visible field quadrilateral.

The resulting metric coordinates are added to the tracklets and can also be used by the linking module to assess the geometric plausibility of candidate associations.

3.2.2 Team and Role Classification

Team membership is estimated from the appearance of the uniform. For each tracklet, crops are extracted from multiple frames and represented using color features in HSV space. Clustering these descriptors separates the two main groups corresponding to the teams on the field.

The classification also considers referees and goalkeepers. Referees are identified through the color compatibility of their uniforms, while goalkeepers are distinguished from outfield players while keeping role and team membership as separate attributes. A confidence value is associated with each estimate, so that an uncertain classification is not necessarily used as a hard constraint.

Classification is performed both before and after linking. The first pass provides information that is useful for tracklet consolidation, while the second exploits the longer trajectories obtained after linking to produce the labels used by subsequent stages.

3.2.3 Tracklet Linking

ByteTrack can fragment the trajectory of the same athlete in the presence of occlusions, exits from the camera's field of view, or temporary association failures. The *tracklet-linking* module therefore attempts to merge distinct fragments when the available evidence indicates that they belong to the same subject.

Linking evaluates pairs of tracklets according to several criteria:

- **temporal and spatial compatibility**, based on the distance between the endpoints of the fragments;
- **appearance similarity**, used as an additional consistency signal;
- **team compatibility**, applied as a constraint when both estimates are reliable;
- **geometric plausibility**, when metric field coordinates are available.

The last criterion compares the displacement implied by two tracklets with the elapsed time, allowing associations that are incompatible with plausible athlete motion to be penalized.

The result of linking is a consolidated trajectory identified by `display_track_id`, which is distinct from the local `raw_track_id` produced by the tracker. Consolidation increases the amount of evidence available to subsequent stages without directly assigning a player identity.

Additional linking can be performed after jersey-number recognition, when a consistent reading provides more identity-specific evidence than appearance similarity or geometric proximity alone.

3.3 Jersey Number Recognition

Jersey number recognition provides the main source of direct evidence for player identification. Unlike the previous stages, which provide contextual information such as position, team, and role, this stage aims to determine which number is actually observable along the tracklet.

The system uses two independent and complementary recognition paths. The first treats the athlete crop as a generic image containing text; the second explicitly localizes the jersey-number region and recognizes its digits using a

specialized model. The two paths do not directly assign an identity: each produces its own evidence, which is kept separate until the evidence and decision layer described in Section 3.4.

Both paths exploit the temporal redundancy of the tracklet. Multiple crops are analyzed independently and the resulting predictions are aggregated, so that a number readable only in a few favorable frames can still provide evidence for the entire trajectory.

3.3.1 Generic OCR Path

The first path applies a generic text-recognition system directly to athlete crops. The implementation uses MMOCR with DBNet for text-region localization and SAR for recognition, with EasyOCR as an alternative when MMOCR is not available. The models are used without football-specific fine-tuning.

All quantitative experiments reported in Chapter 5 use MMOCR with DBNet and SAR. EasyOCR is retained exclusively as an implementation fallback and does not contribute to the reported results.

Since the position of the jersey number within the crop is not known in advance, the system generates multiple candidate views using spatial windows and preprocessing transformations. The resulting readings are then aggregated at tracklet level through weighted voting. The path produces evidence only when the agreement among observations is sufficient; otherwise, it abstains.

Several domain-specific rules are used to reduce spurious readings. In particular, single-digit predictions are treated more cautiously because they may result from partial visibility of a two-digit number. Goalkeepers are also excluded from this path because their visual characteristics differ from those

of outfield players.

The main advantage of this approach is its independence from a specialized jersey-region detector. The path can therefore produce evidence even when explicit jersey-number localization fails. On the other hand, processing multiple views of the same crop increases computational cost and also increases the chance of detecting textual or graphical elements unrelated to the jersey number.

3.3.2 Jersey-Number Region Path

The second path explicitly separates localization from recognition. A dedicated detector from the YOLO family, trained on annotated jersey-number regions, identifies the candidate region within the athlete crop to be passed to the recognizer. When the detector does not identify a valid region, the path abstains and no evidence is produced.

The recognizer is a convolutional recurrent network trained using Connectionist Temporal Classification (CTC). As illustrated in Figure 3.2, feature extraction is performed by a truncated ResNet-18. The resulting feature maps are aggregated along the vertical dimension, producing a sequence of vectors that preserves the horizontal ordering of visual features. The sequence is then processed by a bidirectional LSTM and projected, at each position, onto eleven possible symbols: the ten digits and the CTC *blank* symbol.

The CTC formulation makes it possible to recognize variable-length sequences without requiring an explicit alignment between feature-map positions and the digits of the jersey number. During CTC decoding, consecutive repeated symbols are collapsed and *blank* symbols are removed. The presence of a *blank* between two occurrences of the same digit therefore makes it

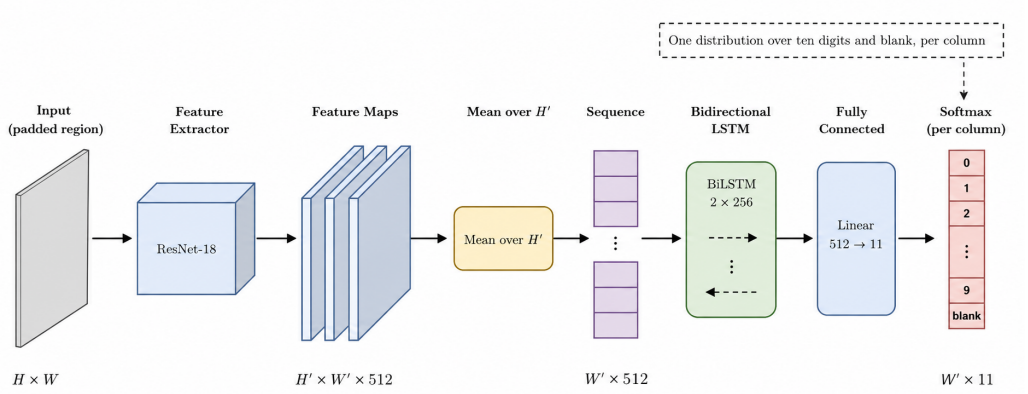


Figure 3.2: Architecture of the CRNN-CTC jersey-number recognizer.

possible to correctly represent numbers such as 11 or 22.

Rather than relying only on greedy decoding, the system evaluates the CTC probability of admissible one-digit and two-digit numbers. For each candidate, all CTC alignments compatible with the corresponding sequence are considered. Its score therefore derives from the total probability of the possible paths rather than from a single alignment. Figure 3.3 illustrates the mechanism using a simplified example, showing one of the paths compatible with the candidate being evaluated.

The output of the recognizer is therefore a ranked list of numeric candidates with their associated scores. The representation preserves both the highest-scoring candidate and lower-scoring alternatives, avoiding the premature removal of uncertainty. This property allows subsequent stages to use external information, such as the match roster, to prefer a compatible candidate that is already present in the ranking without requiring a new inference pass.

The region-based path is more specialized than generic OCR and benefits from the explicit separation between localization and recognition. Its effec-

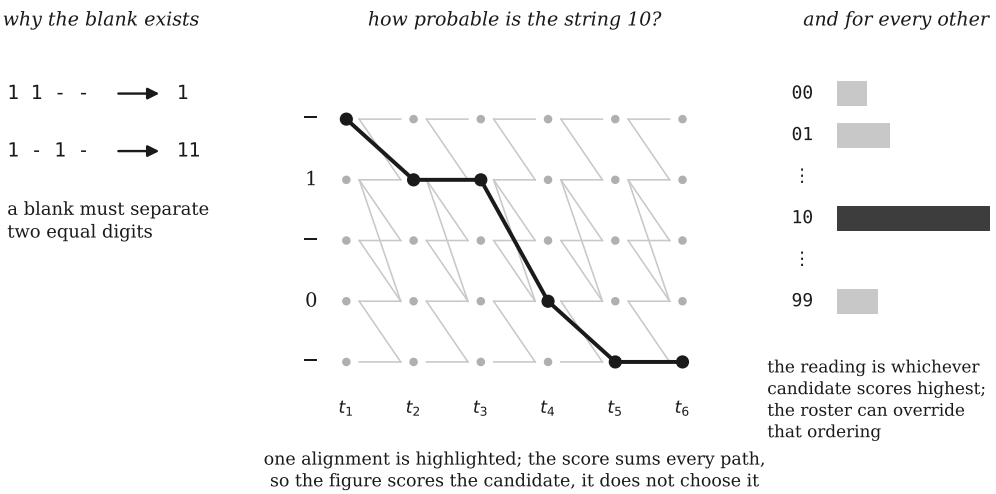


Figure 3.3: Evaluation of a candidate through compatible CTC alignments.

tiveness nevertheless depends on the coverage of the region detector. When the detector does not localize the number, the recognizer receives no input and cannot produce evidence.

The outputs of the two paths therefore remain separate: each can produce a prediction, a ranked list of candidates, or abstain. The evidence is associated with its corresponding source and transferred, without arbitration, to the evidence and decision layer described in Section 3.4.

3.4 Evidence and Decision Layer

The two recognition paths can produce agreeing predictions, conflicting predictions, or abstain. The pipeline therefore separates evidence production from the choice of which jersey number should be used.

The *evidence and decision layer* is the central point of the proposed architecture. The recognizers operate as independent sources, while a configurable policy determines which evidence becomes operational. Arbitration

can therefore be modified without retraining the models or changing the subsequent identity-assignment stage.

3.4.1 Evidence Representation

Recognizer predictions are not written directly into the tracklet's jersey number. Each result is instead stored as a piece of *evidence*, including at least:

- the tracklet to which it refers;
- the proposed number and its associated score;
- the source that produced the prediction.

Evidence from different sources is stored separately. A new observation does not replace existing evidence, and the predictions produced by the two recognizers can therefore be compared even when they disagree. If a recognizer abstains, no evidence is generated. Abstention thus remains distinct from a low-confidence prediction.

3.4.2 Arbitration Between Recognizers

The operational jersey number is selected according to a policy that defines a priority order among the sources. In the final configuration adopted in this work, the ordering is:

```
sources:  [jersey_region_ctc, jersey_ocr_primary]
```

The first source that provides valid evidence determines the operational number. Lower-priority sources are considered only when all higher-priority sources have abstained. The ordering therefore changes the *decision*, but

not the predictions produced by the recognizers, which remain available for analysis.

The behavior can be summarized in three cases:

- **both sources produce a prediction:** the higher-priority source is selected;
- **only one source produces a prediction:** its evidence is used;
- **both sources abstain:** the tracklet remains without an operational jersey number.

The result is therefore an *operational jersey number*, together with the source and score of the selected evidence. Converting this number into a player identity remains the responsibility of the next stage, which also integrates team, role, and roster information.

3.5 Identity Assignment

The jersey number produced by the *evidence and decision layer* does not yet constitute a player identity. The next stage combines this evidence with team, role, and roster information to determine which player can be associated with the tracklet.

The assignment is formulated as a global matching problem between tracklets and players in the roster. In this way, identity is not determined by considering each tracklet independently, but by taking into account the overall compatibility of the assignments and the constraints linking the different players.

3.5.1 Roster-Based Filtering

The match roster provides prior knowledge about the set of possible identities. For each tracklet, the estimated team and role first determine which players in the roster can be considered as candidates.

The operational jersey number produced by the evidence and decision layer is then compared with the numbers present in the roster. When the recognized number corresponds to a player from the assigned team, this information directly restricts the candidate set. Conversely, a number that is absent from the roster is not interpreted as a new identity: the candidate is either removed or assigned a lower compatibility.

The filter can also exploit the ranked list produced by the region-based recognition path. When multiple numeric candidates are available, a lower-scoring value may be preferred over the top-ranked one if it is compatible with the roster, provided that it satisfies the required minimum confidence conditions. The roster therefore acts mainly as a validity constraint, but it can also select among alternatives that have already been produced by the recognizer.

This step does not introduce new jersey-number evidence. It only uses the evidence already produced by the recognizers and combines it with roster knowledge.

3.5.2 Global Assignment

The candidates obtained through roster-based filtering are evaluated using a compatibility cost between each tracklet and each player. The cost combines the main available pieces of evidence, including team, jersey number, role, and, to a lesser extent, information about the trajectory and crop

quality.

The problem is solved globally using the Hungarian algorithm. The solution therefore seeks the set of correspondences that minimizes the overall cost while preserving assignment uniqueness. A player cannot be assigned simultaneously to multiple tracklets within the same solution. Figure 3.4 shows a simplified example of the cost matrix used to represent the problem: rows correspond to tracklets, columns to candidate players, and the highlighted cells identify the correspondences selected by the global solution.

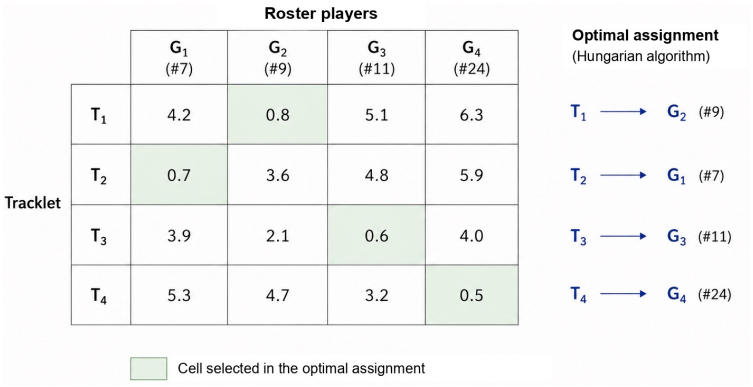


Figure 3.4: Example of global assignment between tracklets and players through cost minimization.

The global formulation is particularly important when multiple tracklets contain similar or incomplete evidence. A locally optimal choice for a single tracklet could assign an identity that is more compatible with another tracklet. Joint optimization makes it possible to resolve these ambiguities by considering all available correspondences simultaneously.

The cost is deliberately designed to remain interpretable rather than being learned. Its terms represent distinct pieces of evidence and can therefore be analyzed separately to understand why a given identity was preferred over the available alternatives.

3.5.3 Constraints and Propagation

The global assignment is finally subject to domain-specific constraints. These include uniqueness of the jersey number within a team and compatibility between the tracklet role and the player role. An assignment that violates a hard constraint is removed, and the corresponding tracklet is left unidentified.

The system is also not required to accept the optimal solution produced by the Hungarian algorithm. An assignment is accepted only if it satisfies a cost threshold and is supported by sufficient positive evidence. If these conditions are not met, the tracklet remains *unknown* rather than being assigned an identity solely because an unused slot remains available in the roster.

Once an identity has been confirmed, it can be propagated across linked tracklets. Propagation allows evidence observed in only one fragment, for example a jersey number readable in a few frames, to extend the identity to the entire consolidated trajectory. Tracklet linking therefore becomes an integral part of identification: an incorrect link can propagate a correct identity to a different athlete, while a correct link allows a single reliable observation to be exploited over a longer temporal interval.

3.6 Output Representation

All stages of the pipeline share a tabular representation indexed by `(frame_id, track_id)`. Starting from the initial detections, each stage adds the information it produces without overwriting the information generated by previous stages. The same structure therefore makes it possible to follow the evolution of an observation from the initial detection to the eventual identity

assignment.

For each tracklet, the representation stores, when available, detection geometry, the raw and consolidated identifiers, team, role, projected field position, the evidence produced by the recognizers, and the jersey number selected by the evidence and decision layer. The final identity is added separately and may remain unassigned when the available evidence is insufficient.

The representation therefore preserves the distinction between *observation* $\rightarrow$ *evidence* $\rightarrow$ *decision* $\rightarrow$ *identity*. This distinction is particularly important for the experimental analysis: an incorrect prediction can be traced back to the recognizer that produced the evidence, to the policy that selected it, or to the assignment stage that converted it into an identity.

In addition to the tabular output, the pipeline stores the diagnostic artifacts produced by the individual stages, including the configurations used and the evidence log. These artifacts are not required to use the final output, but they make it possible to reconstruct intermediate decisions and compare configurations that differ in a single component.

This structure provides the basis for the experiments in Chapter 5, where the different pipeline components are evaluated both individually and in their end-to-end behavior.

Chapter 4

Experimental Methodology

The experimental evaluation of the system considers both individual components and the complete pipeline. The experiments are organized so as to isolate, when necessary, the contribution of individual modules and then evaluate their effect under the operational conditions of the full pipeline.

The protocol defines the data used for training and evaluation, the evaluation regimes under which the experiments are conducted, the adopted metrics, and the statistical procedure used to compare different configurations. The choices concerning the models and the execution setup are reported at the end of the chapter.

4.1 Datasets and Data Splits

The experimental evaluation mainly relies on SoccerNet Game State Reconstruction (SoccerNet-GSR) [22], described in Section 2.1. The dataset contains 200 broadcast sequences, of which 164 are annotated and divided into 57 training sequences, 58 validation sequences, and 49 test sequences. The remaining 36 sequences form the challenge set and are not used in eval-

uations that require *ground truth*.

The training split is further divided into disjoint subsets:

- **37 sequences** for training the jersey-number recognition components;
- **9 sequences** for checkpoint selection and the choice of operational thresholds;
- **11 sequences** for verifying the final configuration, with no further changes to models, checkpoints, or thresholds.

The validation split is used for intermediate evaluations. A block of 12 sequences is used to evaluate detection and tracking and for the related ablations, while 7 sequences are used to compare the decision policies. These latter sequences are selected automatically by requiring at least eight player or goalkeeper tracklets with an annotated jersey number, without using model predictions as a selection criterion. The block contains 154 tracklets in total, of which 116 have an annotated jersey number. These 116 tracklets constitute the evaluation units, while the seven sequences serve as the sampling units for the statistical analysis.

The test split is used for the final evaluations. The complete set of 49 sequences is used to compare the jersey-number recognizers, while a random subset of 12 sequences, sampled using a recorded seed, is used for the end-to-end evaluation. This latter block is not used to select the arbitration policy, modify thresholds, or choose checkpoints.

4.2 Evaluation Regimes

4.2.1 Offline Evaluation

In the *offline* regime, detection and tracking information is provided directly by the *ground truth*. Bounding boxes and the grouping of detections into tracklets are therefore treated as correct by construction, allowing subsequent pipeline components to be evaluated in isolation, particularly jersey-number recognition and the arbitration layer.

This regime enables a controlled comparison because all configurations receive the same tracklets and the same crops. The observed differences can therefore be attributed to the components under evaluation without the direct influence of detection or tracking errors. For jersey-number recognition, the experiment uses bounding boxes and tracklet identities provided by the *ground truth*. In the arbitration evaluation, the same offline setting is applied to the 116 tracklets with an annotated jersey number in order to compare the decision policies.

The offline regime nevertheless represents an idealized condition. Crops derived from the *ground truth* are generally more favorable than those available during execution of the complete pipeline. Results obtained under this regime should therefore be interpreted as a measure of the potential of the component rather than as an estimate of end-to-end performance.

4.2.2 Pipeline Evaluation

In the *pipeline* regime, the system is executed using the complete configuration described in Chapter 3. Detection, tracking, localization, and classification therefore produce the actual inputs used by the subsequent stages, without any intervention from the *ground truth* during processing.

The *ground truth* is used only during the final evaluation stage. Predictions are matched to annotated athletes using bounding-box overlap, allowing the effect of the evaluated modifications to be measured under the actual operational conditions of the system.

The comparison between the offline and pipeline regimes therefore distinguishes improvements observed under controlled conditions from those that remain when the system is executed end to end. Results from the two regimes are kept separate.

4.2.3 Out-of-Domain Evaluation

Broadcast recordings external to SoccerNet-GSR are also used to evaluate the behavior of the system outside the dataset domain. These videos do not provide complete positional annotations and therefore do not allow the *ground-truth*-based metrics used on SoccerNet-GSR sequences to be computed directly.

Evaluation on these recordings consequently focuses on *coverage*, on the availability of the different sources, and on agreement or disagreement between them, all of which can be measured without positional annotations. Only a small subset of Inter–Atalanta, consisting of 36 tracklets with manually verified identities, also allows identity accuracy to be estimated. Accuracy is not reported for the other clips. The analysis instead concerns the availability of evidence and the behavior of the different sources under different domain conditions, without directly comparing these results with the metrics obtained on the annotated dataset.

4.2.4 Oracle Substitution

Oracle ablation is applied after inference to the artifacts already produced by the pipeline. For each stage, the actual output is replaced with the corresponding *ground truth*, without repeating inference and without propagating the replacement backwards to previous stages.

Two complementary directions are considered. In the first, a single stage is made oracle while all other stages remain unchanged, allowing the marginal gain relative to the current system to be measured. In the second, a single stage remains unchanged while all other stages are made oracle, allowing the isolated cost of the stage under consideration to be measured relative to the ideal limit. Annotations that do not contain the field coordinates required to compute GS-HOTA are excluded from the oracle evaluation.

When all the considered stages are replaced with *ground truth*, the *all-oracle* configuration must return a GS-HOTA value of 1.0. This serves as a consistency check for both the substitution procedure and the metric implementation.

4.3 Metrics

The metrics used depend on the level of the pipeline under evaluation. Standard metrics are adopted for detection and tracking, while jersey-number recognition requires measures that distinguish the system's ability to produce an output from the correctness of that output. Finally, identity evaluation considers the correctness of the final identity assigned to each tracklet.

4.3.1 Detection and Tracking

Detection performance is evaluated using *recall* and *Average Precision* (AP), while tracking is evaluated using HOTA [17]. HOTA provides a joint measure of detection and association performance and is also analyzed through its Detection Accuracy (DetA) and Association Accuracy (AssA) components.

In addition to these aggregate metrics, *identity switches* and fragmentations are considered in order to characterize association errors between consecutive frames. The metrics are computed on the sequences belonging to the corresponding evaluation block.

4.3.2 Jersey Number Recognition

The evaluation of jersey-number recognition distinguishes tracklets with an annotated number from those for which the number is not visible. This distinction is necessary because the absence of a prediction cannot be interpreted in the same way when the number is observable and when no reference number is available.

Three quantities are considered for tracklets with an annotated jersey number. *Coverage* measures the proportion of tracklets for which the system produces a number, while *accuracy on emitted* measures the proportion of emitted predictions that match the annotated number. *Overall accuracy* instead considers all tracklets with an annotated jersey number and measures the proportion for which the correct number is assigned.

The three quantities are related by

$$\text{Overall Accuracy} = \text{Coverage} \times \text{Accuracy on Emitted}.$$

This distinction is particularly important when comparing configurations with different tendencies to emit predictions. A system can achieve high accuracy on its emitted outputs by restricting predictions to the most confident cases, without necessarily increasing the overall number of correctly recognized tracklets.

For tracklets without a visible jersey number, the *false emission rate* is considered instead. It is defined as the proportion of tracklets for which the system nevertheless produces a number. Such emissions are considered incorrect by definition and are therefore kept separate from the metrics computed on tracklets with an annotated jersey number.

4.3.3 Identity Accuracy on Manual Annotations

When the evaluation concerns identity assignment, a tracklet is considered correct if the identity assigned by the system matches the available reference label for that sample.

This measure is used primarily for the small manually annotated roster benchmark on recordings external to SoccerNet-GSR, for which complete positional *ground truth* is not available. Results obtained on these data are therefore reported separately from the metrics computed on the annotated dataset sequences.

4.3.4 GS-HOTA

GS-HOTA [22] extends tracking evaluation by combining localization, temporal association, and identity attributes. The similarity between a prediction P and an annotation G is the product of a spatial component and an identity component:

$$\mathrm{Sim}_{\mathrm{GS}}(P, G) = \mathrm{LocSim}(P, G) \cdot \mathrm{IdSim}(P, G).$$

The spatial component depends on the Euclidean distance between metric field positions:

$$\mathrm{LocSim}(P, G) = \exp\left(\ln(0.05)\frac{\|P - G\|_2^2}{\tau^2}\right), \qquad \tau = 5\,\mathrm{m}.$$

For players and goalkeepers, the identity component requires team, role, and jersey number to be jointly correct. Attributes that are not annotated are ignored. Because the formulation is multiplicative, a single incorrect attribute makes the identity similarity equal to zero even when localization and tracking are correct. A missing attribute that is required by the *ground truth* therefore makes the prediction ineligible for matching with that subject.

4.3.5 Transition Analysis

When comparing two configurations, the analysis considers not only the difference between aggregate accuracies, but also how the outcome changes for individual tracklets. Since the two configurations are applied to the same elements, the comparison is paired, and each tracklet can be classified according to the transition observed between the two configurations.

The main transitions considered are:

- **Recovery:** a transition to a correct result from either a previously incorrect emission or an abstention;
- **Regression:** the loss of a previously correct prediction;
- **New wrong emission:** a transition from abstention to an incorrect emission.

These quantities make it possible to distinguish the benefit introduced by a new configuration from the cost required to obtain it. In particular, higher coverage is not automatically considered beneficial: newly emitted predictions must also be evaluated in terms of correctness.

4.4 Statistical Analysis

Comparisons between configurations are performed on the same evaluation elements, resulting in paired comparisons. To quantify uncertainty, resampling is performed at sequence level rather than at individual tracklet level. Tracklets belonging to the same sequence share recording conditions, illumination, and game context and therefore cannot be treated as completely independent observations.

4.4.1 Uncertainty Estimation

Confidence intervals are estimated using sequence-level bootstrap. For each evaluation block, 10 000 bootstrap replicates are generated by sampling, with replacement, the same number of sequences as in the original block. For each replicate, the statistic of interest is recomputed, in particular the difference between configurations based on the macro-average of the sampled sequences. A fixed and recorded seed is used for all replicates to ensure reproducibility. The reported interval is the 95% percentile interval of the bootstrap distribution.

The estimates reported in Chapter 5 are accompanied by their corresponding confidence intervals. The procedure therefore treats sequences, rather than individual tracklets, as the independent sampling units, taking the hierarchical structure of the data into account.

4.4.2 Paired Comparisons and Significance Tests

When two configurations are compared, each tracklet provides a pair of outcomes. The exact McNemar test, equivalent to an exact binomial test on the discordant pairs only, is used as a complementary analysis of the direction of the transitions between configurations.

The test considers only discordant observations and, under the null hypothesis, assumes that the two possible types of discordance have equal probability. Since tracklets belonging to the same sequence share recording conditions and cannot be considered completely independent, the primary inference concerning the magnitude and uncertainty of the effect relies on bootstrap confidence intervals obtained through sequence-level resampling.

The McNemar test is therefore reported together with the observed changes and confidence intervals, rather than replacing the sequence-level analysis. The more detailed transitions defined in Section 4.3.5, such as *recovery*, *regression*, and *new wrong emission*, are instead used to describe the nature of the observed changes.

For continuous metrics aggregated at sequence level, the direction of the effect is also assessed using a sign test on the paired differences, excluding sequences with tied results.

4.5 Experimental Configuration

The experiments were conducted using configurations fixed before evaluation. In A/B comparisons, only one variable is changed at a time. Checkpoints, inputs, seeds, front-end, calibration, and shared modules remain unchanged when the comparison is intended to isolate the effect of the arbitration policy. In this way, each comparison measures the effect of the

modification under consideration without introducing changes in other parts of the system.

4.5.1 Model Configuration

The components trained or adapted in this work include the person detector, which is fine-tuned on SoccerNet-GSR, the jersey-number region detector, and the CTC recognizer. The region detector and the CTC recognizer use the same 37 sequences from the training subset dedicated to recognizer training, with a fixed seed. The person detector uses the checkpoint obtained through fine-tuning on SoccerNet-GSR described in Section 3.1.

Table 4.1: Configuration of the jersey-number recognition components trained in this work.

	Region detector	CTC recognizer
Architecture	YOLO, *small* variant	ResNet-18 with BiLSTM
Training sequences	37	37
Seed	20260719	20260719

The comparison between region-detector configurations uses checkpoints obtained from the same set of annotated images and with the same seed. The configurations differ in training duration and, in one case, in initialization through external pre-training on the dataset by Li et al. [13].

4.5.2 Execution Environment

The experiments were conducted on a single workstation running Ubuntu 24.04.4 LTS, equipped with a 6-core, 12-thread Intel Core i5-10600KF processor, 31 GiB of RAM, and an NVIDIA GeForce RTX 3060 GPU with 12 GiB

of video memory. No distributed or multi-GPU configurations were used.

The different components are executed in isolated software environments due to the different dependencies required by the frameworks involved. The pipeline stages communicate through the tabular artifacts defined in Section 3.6. The versions of the main dependencies and the configurations required to reproduce the experiments are fixed in the project repository.

4.5.3 Computational Cost

The computational cost of recognition depends on the number of crops analyzed for each tracklet. A larger crop budget increases inference cost, but can also provide more observations to the recognizers. This budget is therefore treated as an experimental variable, and its effect on coverage is evaluated in Section 5.4.3.

Computational cost is analyzed primarily through the number of processed crops and, in the comparison between arbitration policies, through a controlled runtime measurement on the same sequence. A systematic benchmark of latency or FPS for the complete pipeline is not provided. In A/B comparisons, the crop budget and all other parameters that are not the subject of the experiment are kept unchanged.

Chapter 5

Experimental Results

This chapter experimentally analyzes the behavior of the pipeline, proceeding from the upstream components to the end-to-end evaluation. The objective is not only to quantify the performance of individual modules, but also to identify which errors propagate through the pipeline, which interventions produce a measurable improvement, and which instead act on a stage that is not the actual limiting factor.

The datasets, evaluation regimes, metrics, and statistical procedures used in the experiments are defined in Chapter 4. In particular, the results are distinguished between offline evaluations, in which some inputs are provided by the *ground truth*, evaluations of the complete pipeline, and analyses on footage outside the SoccerNet-GSR domain. Metric definitions are therefore not repeated in the following sections. For each experiment, the comparison being performed, the observed result, and its implication for subsequent stages are specified instead.

The analysis follows the causal order of the pipeline. Detection and tracking are first examined to determine whether they produce sufficiently reliable tracklets and whether calibration provides usable metric coordinates. A sub-

sequent oracle ablation then quantifies how much each stage contributes to the gap from ideal performance. Starting from this diagnosis, jersey-number recognition and the factors limiting its coverage are analyzed, followed by the policy used to combine the different pieces of evidence. Finally, the resulting configuration is evaluated out of domain and within the complete system using GS-HOTA, with a final verification on a test block that was not used to select the arbitration policy or the thresholds of the end-to-end configuration.

5.1 Detection and Tracking

The first requirement of the pipeline is that the front-end produce sufficiently complete observations and, most importantly, temporally coherent tracklets. An error at this stage cannot be fully recovered by the identification modules: fragmentation distributes the evidence associated with the same athlete across multiple tracklets, while an identity switch can combine observations belonging to different players.

The front-end was evaluated on the validation pilot consisting of twelve sequences, for a total of 9000 frames. All downstream identity modules were disabled during this evaluation. Detection metrics are computed on the raw detector output, before the filtering introduced by the tracker, while tracking metrics are computed on the identities actually provided to the subsequent stages of the pipeline.

5.1.1 Front-End Quality

The detector achieves high recall: at IoU 0.50, 95.8% of the annotated athletes are recovered and AP reaches 0.951. The same metric, however,

Table 5.1: Detection and tracking performance on the twelve-sequence validation pilot (9000 frames). DetA, AssA, and HOTA are averaged over the nineteen evaluation thresholds; MOTA is computed at IoU 0.50. Identity switches and fragmentations are aggregate counts over the twelve sequences.

Detection		*Tracking*	
Recall @ 0.50	95.8%	DetA	0.710
AP @ 0.50	0.951	AssA	0.517
AP @ 0.75	0.619	HOTA	0.602
		MOTA	0.872
		ID switches	1298
		Fragmentations	1941

drops to 0.619 when the overlap threshold is increased to 0.75. The main issue is therefore not the absence of players from the detections, but the accuracy with which their bounding boxes are localized.

This distinction is relevant for the tracker adopted in the pipeline. ByteTrack associates detections between consecutive frames mainly on the basis of geometry and box overlap, so localization errors can propagate directly to the temporal association stage.

The same phenomenon is reflected in the HOTA components. DetA reaches 0.710, whereas AssA remains at 0.517. This gap indicates that the system is considerably more effective at determining *which* athletes are present than at maintaining a temporally consistent identity for their observations. The 1298 identity switches and 1941 fragmentations observed at the output of the front-end further confirm the weakness of temporal consistency. The next subsection separates the contribution of raw tracking from that introduced by linking.

MOTA, equal to 0.872, instead provides an apparently more favorable picture. This result does not contradict HOTA. With high recall, false positives and false negatives have a relatively limited impact, while association errors affect MOTA less strongly than a metric explicitly designed to separate detection and association. For this pipeline, MOTA alone would therefore hide a substantial part of the observed problem.

Overall, the front-end detects athletes reliably, but the temporal consistency of the tracklets remains significantly weaker than detection quality. This result motivates the analysis of tracklet linking, which was introduced specifically to reconnect fragments that the tracker is unable to keep together.

5.1.2 Effect of Tracklet Linking

The linker described in Section 3.2.3 operates downstream of the tracker and attempts to reconnect fragments assigned to the same athlete, using the admissibility constraints available in the pipeline. Since the execution preserves both the raw identity produced by the tracker and the identity obtained after linking, its effect can be measured directly through a paired comparison on the same twelve sequences.

Tracklet linking does not produce a measurable improvement in tracking performance. For HOTA, DetA, and AssA, the confidence intervals include zero, and the sequences are distributed relatively evenly between improvements and degradations. MOTA instead shows a small but consistent negative change: none of the sequences improves and eleven out of twelve worsen.

The count-based metrics help interpret the result. If the linker primarily reconnected fragments belonging to the same athlete, a reduction in identity switches would be expected. The opposite is observed: identity switches increase from 1139 to 1298, corresponding to 159 additional events, while

Table 5.2: Effect of tracklet linking on the validation pilot. Differences are expressed as linking minus raw tracking, with 95% confidence intervals obtained using sequence-level bootstrap. I/W indicates the number of improved/worsened sequences.

	Raw	Linking	Δ (95% CI)	I/W
HOTA	0.6041	0.6018	−0.0022 [−0.0132, +0.0101]	5/7
DetA	0.7103	0.7101	−0.0002 [−0.0008, +0.0004]	4/8
AssA	0.5189	0.5167	−0.0022 [−0.0200, +0.0193]	5/7
MOTA	0.8728	0.8718	−0.0010 [−0.0014, −0.0006]	0/11
ID switches	1139	1298	+159	–
Fragmentations	1939	1941	+2	–

the number of fragmentations remains essentially unchanged. This result is consistent with incorrect merges in which fragments belonging to different athletes are linked, rather than with the systematic recovery of correct fragmentations.

It follows that, under the configuration considered, the limitation of the linker does not appear to be its ability to merge tracklets more aggressively, but rather the quality of the proposed associations. The current admissibility constraints are not sufficient to guarantee that a link is established between observations of the same athlete.

This conclusion concerns tracking metrics and does not imply that the linker is necessarily useless for every downstream stage. A longer tracklet can provide more crops for jersey-number voting, a benefit that HOTA and MOTA do not measure directly. The available data do not, however, support the hypothesis that linking improves the temporal consistency of identities in the current configuration.

The result therefore directs the subsequent analysis in two directions. First, it is necessary to determine whether geometric information on the field can make associations between fragments more selective. Second, before attributing the main limitation of the pipeline to the linker, the contribution of the other components must be quantified through a controlled end-to-end evaluation. The first question is addressed in the following section through the calibration analysis, while the second is revisited in the oracle ablation.

5.2 Camera Calibration

The previous section showed that the main limitation of the front-end does not lie in its ability to detect athletes, but in the temporal consistency of the tracklets. The pipeline nevertheless has access to an additional source of information that may contribute to association: the metric position of athletes on the playing field. Before using this information as a geometric constraint, it is necessary to verify that calibration is sufficiently accurate, since an error of several meters would make any criterion based on motion plausibility of limited value.

The pipeline supports two calibration sources, described in Section 3.2.1: an estimate based on the visible field quadrilateral and TVCalib. Table 5.3 compares their localization error with respect to the *ground-truth* metric coordinates on the same twelve-sequence validation pilot used for the front-end analysis.

The difference between the two sources is substantial. The field-quadrilateral estimate produces a median error of 35.78 m and a 90th percentile of 58.28 m. Errors of this magnitude are too large for field position to be used as reliable metric information. On a field that is 105 m long, a deviation of several tens

Table 5.3: Athlete position error on the field for the twelve-sequence validation pilot, measured with respect to the metric coordinates of the *ground truth*. All values are expressed in meters.

Source	Median	Mean	P90
Field quadrilateral	35.78	–	58.28
TVCalib	**1.10**	1.43	**2.71**

of meters can place an athlete in a completely different region of the playing field.

TVCalib instead reduces the median error to 1.10 m, with a mean of 1.43 m and a 90th percentile of 2.71 m. The resulting localization is therefore sufficiently accurate to justify the use of geometric constraints based on motion, although their effectiveness is evaluated separately in the following experiments.

For this reason, all results in this chapter that require positions on the playing field are computed using TVCalib. The field-quadrilateral estimate is retained in the pipeline only as a fallback, so that the absence of a calibration produced by the model does not prevent subsequent stages from being executed.

5.2.1 Geometry Does Not Solve Tracklet Linking

The availability of accurate metric coordinates makes it possible to test a hypothesis suggested by the results in Section 5.1: linker errors might be reduced by removing or penalizing associations that require physically implausible displacements.

The two geometric mechanisms based on the metric coordinates described in Section 3.2.1 were evaluated by changing one variable at a time. The first

applies a hard speed constraint, excluding associations that are incompatible with athlete motion. The second uses the same information as a soft re-ranking term for the candidate associations. Both mechanisms were evaluated end to end on the twelve-sequence validation pilot.

Neither intervention produces an improvement. The hard filter changes HOTA by -0.00076 and improves only one sequence out of twelve, while soft re-ranking produces a change of -0.00019 without improving any sequence. Relative to a HOTA value of approximately 0.602, both effects are smaller than one thousandth and provide no evidence of a benefit from the geometric constraint.

The negative result does not imply that field position is too inaccurate to be useful. The previous experiment shows that TVCalib achieves an error on the order of a few meters. To identify the cause of the lack of improvement, the list of candidate merges actually considered by the linker was therefore analyzed.

Among 50 cases for which association correctness could be verified, the correct link was entirely absent from the available candidate set in 22 cases. In these cases, the problem therefore occurs before the stage affected by the geometric constraint. The linker is not choosing the correct candidate from among several alternatives because the correct candidate was never generated.

This result explains why neither hard filtering nor re-ranking can produce a substantial improvement. Both can modify the selection among already available candidates, but neither can recover an association that is absent from the initial set. The observed limitation therefore lies in *candidate generation*, upstream of the ranking stage, rather than in the absence of a sufficiently informative geometric criterion.

The experiment thus refines the diagnosis obtained in Section 5.1. The front-end has an association problem, but this problem is solved neither by linking in its current configuration nor by introducing accurate metric geometry into the link-selection stage. Before further modifying a single component, it is therefore necessary to determine how much this limitation contributes relative to the other errors in the pipeline. The following section addresses this question through an oracle ablation of the main stages of the system.

5.3 Ablation Analysis

The previous sections identified two limitations of the front-end: the temporal consistency of the tracklets is considerably weaker than detection quality, and linking does not correct this problem systematically, even when accurate metric coordinates are available. These results do not, however, establish how much these errors contribute relative to those introduced by later stages of the pipeline.

To separate the contributions of the different stages, an oracle ablation is therefore performed following the methodology introduced by the SoccerNet-GSR benchmark. The idea is to replace, during evaluation, the output of one or more components with the corresponding *ground truth* and then recompute GS-HOTA. This makes it possible to measure how much of the gap from ideal performance is associated with the errors of each stage.

The analysis considers five components: detection and tracking, calibration, team, role, and jersey number. Each component is evaluated through two complementary substitutions. In the first case, only one stage is made perfect while all others remain unchanged; the resulting variation measures

the *marginal gain* that can be obtained by correcting that stage starting from the current system. In the second case, all stages except the one under consideration are made perfect; the distance from the fully oracle limit instead measures its *isolated cost*.

The substitutions are applied to the artifacts already produced by the pipeline on the twelve-sequence validation pilot and do not require a new inference pass. *Ground-truth* annotations without the field coordinates required by GS-HOTA are excluded together with the predictions associated with them, since the metric cannot evaluate their spatial component. This operation changes the starting value only marginally: arm A moves from 0.2097 to 0.2099, while arm B moves from 0.2870 to 0.2871.

As a consistency check, when all evaluated components are replaced with the *ground truth*, GS-HOTA reaches exactly 1.0 on all twelve sequences for both arms.

The complete results are summarized in the attribution plot below, while the corresponding point estimates are discussed in the text.

5.3.1 Cost Attribution

The right-hand side of the ablation makes it possible to quantify the isolated cost of the different stages. For arm B, which is used as the reference for attribution, the resulting values are shown in Figure 5.1.

The jersey number has the highest isolated cost, equal to 0.2814 GS-HOTA points. Detection and tracking follow with 0.2566, and calibration with 0.2060. Team and role contribute 0.1809 and 0.1610, respectively. The confidence intervals of the five estimates overlap substantially, so the data do not support the existence of a single dominant bottleneck.

The jersey number therefore has the highest point estimate of isolated

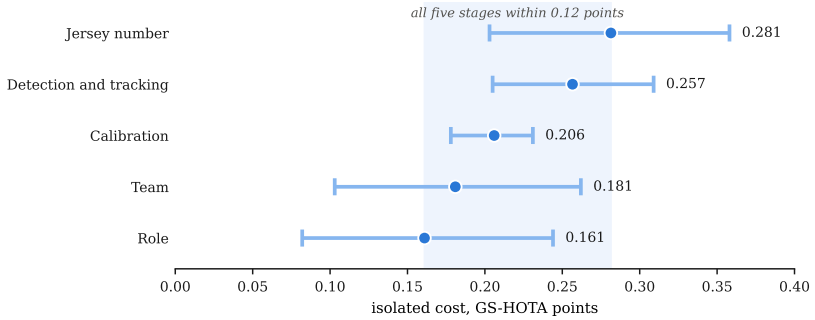


Figure 5.1: Isolated cost of the five stages considered for arm B on the twelve-sequence validation pilot, with 95% sequence-level bootstrap intervals. The cost is measured as the distance from the corresponding all-oracle limit.

cost, but its confidence interval overlaps substantially with those of detection and tracking and calibration. The results consequently do not support attributing the gap from ideal performance to a single dominant component.

The isolated cost should also not be interpreted as the improvement that would be obtained by making that stage perfect in the current system. For example, making the jersey number perfect when all other stages are already oracle is worth 0.2814 points, whereas replacing only the jersey number in the actual system produces a gain of 0.0975. The difference is caused by interactions between errors: a prediction may be penalized by several components at the same time, and correcting only one of them may not be sufficient to recover it completely.

For the same reason, isolated costs are not additive. Their sum is 1.086, which is greater than the overall distance of 0.713 between the actual arm B and the unitary limit. The contributions of the different stages therefore overlap and cannot be interpreted as a linear decomposition of the total error.

Particular care is also required when interpreting calibration. As shown in Section 5.2, TVCalib achieves a mean error of 1.43 m; the cost observed in the ablation therefore does not imply that calibration is qualitatively poor.

Part of this cost results from the sensitivity of the spatial component of GS-HOTA even to relatively small metric errors.

5.3.2 The Role of the Jersey Number

The ablation also makes it possible to isolate the component responsible for the difference between the two decision arms considered in the experiments. The two arms share detection, tracking, calibration, team classification, and role classification, and differ only in the way jersey-number evidence is handled.

The ablation shows a clear separation between configurations in which the jersey number remains unchanged and those in which it is replaced with the *ground truth*. When the jersey number remains unchanged, the gap between the arms ranges from 0.0717 to 0.1084 GS-HOTA points. When the jersey number is instead made oracle, the gap decreases to values between 0.0024 and 0.0031.

In particular, when only the jersey number is made perfect, the two arms reach 0.3819 and 0.3846, respectively, with a difference of only 0.0027. Conversely, when all other components are made perfect and the jersey number remains unchanged, the gap increases to 0.0982.

The near disappearance of the difference when the jersey number is oracle shows that the advantage of one arm over the other is attributable specifically to the handling of this evidence, rather than to differences introduced by the shared stages of the pipeline. The small residual difference, on the order of 0.003, is compatible with differences introduced by the matching procedure used to apply the oracle substitution and compute GS-HOTA, and does not represent a substantial difference between the two systems.

This result also serves as an internal consistency check for the ablation:

when the only component on which the two arms differ is removed, their performance becomes essentially indistinguishable.

5.3.3 Scope of Validity

The ablation identifies the main contributions to the observed error, but it does not simulate the execution of a pipeline in which one of the modules is actually perfect. The substitutions with the *ground truth* are applied after inference. The interactions that a correct prediction would have produced in subsequent stages are therefore not replayed.

The values should consequently be interpreted as a measure of how strongly the error of each stage affects the observed output, rather than as a direct estimate of the improvement that would be obtained by replacing that component with a perfect model.

A second limitation concerns the evaluated surface. GS-HOTA considers field position and the attributes of team, role, and jersey number, but it does not directly measure the final assignment of a player's name from the roster. The effect of the constraints used in the final identification stage must therefore be evaluated separately.

Finally, this attribution is obtained on the twelve-sequence validation pilot. The point estimates, particularly those associated with team and role, should not be interpreted as universal properties of the pipeline. The later evaluation on the test block shows that these contributions depend on how frequently the corresponding modules abstain. The most robust conclusion, shared by both blocks, is that the jersey number, detection and tracking, and calibration contribute on a comparable scale, with no single component dominating the overall error.

The ablation therefore provides the rationale for the following experi-

ments. The front-end remains an important source of error, but Sections 5.1 and 5.2 have already shown that neither linking nor its geometric constraint produces a measurable improvement under the available configuration. The jersey number instead has the highest isolated cost and represents a component that can be modified directly. The following section therefore analyzes the path that produces this evidence, separating recognizer quality from the quantity and quality of the inputs that actually reach it.

5.4 Jersey Number Recognition

The ablation in Section 5.3 identifies the jersey number as one of the main sources of error in the pipeline. The module that produces this evidence, however, involves two distinct problems: locating a region containing the number within the image and correctly recognizing the digits inside it.

This distinction separates two possible limitations. The recognizer may receive a valid region but interpret it incorrectly, or it may receive no input because the region detector fails to localize the number. In the first case, the recognition model must be improved; in the second, increasing recognition accuracy cannot recover the missing information.

The following experiments analyze these two phenomena separately. First, the effect of external pre-training on the recognizer's ability to interpret the same regions is measured. The model is then kept fixed while the stages that determine which inputs actually reach it are analyzed.

5.4.1 Effect of External Pre-training

The CTC recognizer is trained on a relatively limited number of annotated crops from SoccerNet-GSR. To determine whether a greater variety of exam-

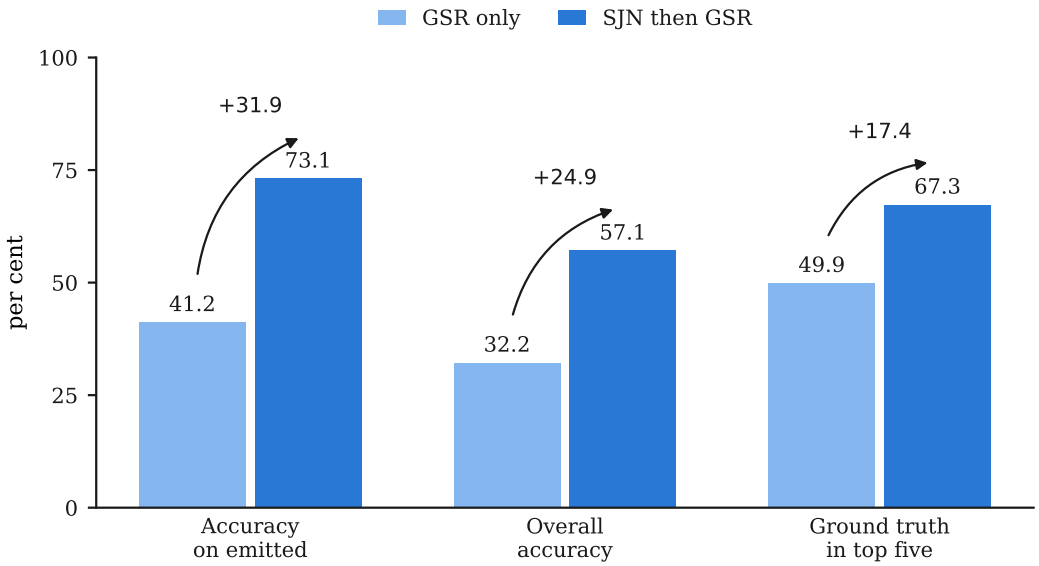


Figure 5.2: Effect of external pre-training on the performance of the CRNN-CTC recognizer.

ples can improve digit recognition, two checkpoints of the same CRNN-CTC architecture are compared. The first is trained exclusively on SoccerNet-GSR data, while the second is pre-trained on the external jersey-number corpus described in Chapter 2 and subsequently fine-tuned on the same SoccerNet-GSR set.

The comparison changes only the initialization of the recognizer. The two checkpoints receive the same crops and the same regions produced by the same frozen detector. In total, the detector identifies 1803 regions from 2954 crops, so the two recognizers are evaluated on exactly the same surface. The experiment uses the frozen test split, which in this case contains 774 tracklets with an annotated jersey number, with bounding boxes and track identities provided by the *ground truth*.

External pre-training produces a clear improvement. The number of correctly recognized tracklets increases from 249 to 442, while accuracy on emitted predictions rises from 41.2% to 73.1%. When tracklets on which the sys-

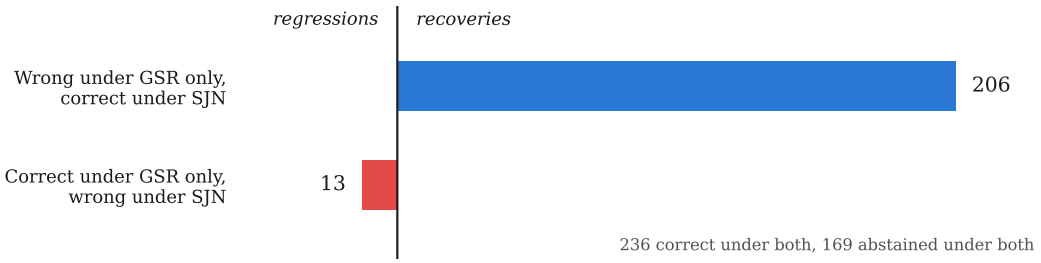


Figure 5.3: Paired transitions between the recognizer trained exclusively on SoccerNet-GSR and the recognizer pre-trained on the external corpus before fine-tuning on the target domain.

tem abstains are also considered, overall accuracy increases from 32.2% to 57.1%, corresponding to a gain of 24.9 percentage points. The main effects of pre-training are summarized in Figure 5.2.

Since both checkpoints operate on exactly the same regions, the difference cannot be attributed to coverage. Pre-training therefore changes what the model is able to read once an input is available, rather than how frequently such an input is provided.

The effect is also visible in the candidate set produced by the recognizer. The correct number appears among the five highest-scoring predictions for 49.9% of the tracklets with the model trained only on GSR, and for 67.3% with the pre-trained model. The benefit therefore does not consist only of reordering candidates that were already present. In 135 tracklets, pre-training brings the correct answer into the top five when it was previously absent.

Figure 5.3 shows the paired transitions between the two checkpoints.

Among the tracklets recognized incorrectly by the model trained exclusively on GSR, 206 are recovered by the pre-trained model. The opposite transition occurs for only 13 tracklets that were previously correct. The exact test on discordant observations gives $p = 7.5 \times 10^{-46}$, while the sequence-level

bootstrap estimates the difference in overall accuracy as +0.249, with a 95% confidence interval of $[+0.222, +0.278]$.

External pre-training is therefore retained in the subsequent configuration. The result also reveals a limitation that is independent of the recognizer: both checkpoints can operate only on the regions provided to them. Improving digit recognition therefore does not increase the amount of available evidence when the jersey number is not localized.

5.4.2 Region Detector Coverage

The region detector is the stage immediately preceding the recognizer. On the split used in the previous experiment, it produces a usable region for 1803 of the 2954 input crops, corresponding to a per-crop coverage of 61.0%. Any crop on which the detector abstains is therefore, by construction, unavailable to the CTC recognizer.

To determine whether this limitation can be reduced simply by extending or transferring training, three YOLO checkpoints are compared. They are obtained from the same 435 annotated images from 37 sequences and use the same seed. The checkpoints use the same 435 annotated images from 37 sequences and the same seed, but differ in training duration and, for the third configuration, in initialization: 10 epochs, 43 epochs, and 25 epochs with initialization from external pre-training.

Figure 5.4 compares the coverage obtained by the three configurations.

At the operational confidence threshold, coverage increases from 60.7% to 62.8% and finally to 63.7%. Considered in isolation, this variation could suggest a progressive improvement. The behavior changes, however, when the threshold is lowered to 0.05: coverage decreases from 75.0% to 72.8% and then to 71.9%. At the same time, the number of tracklets for which no

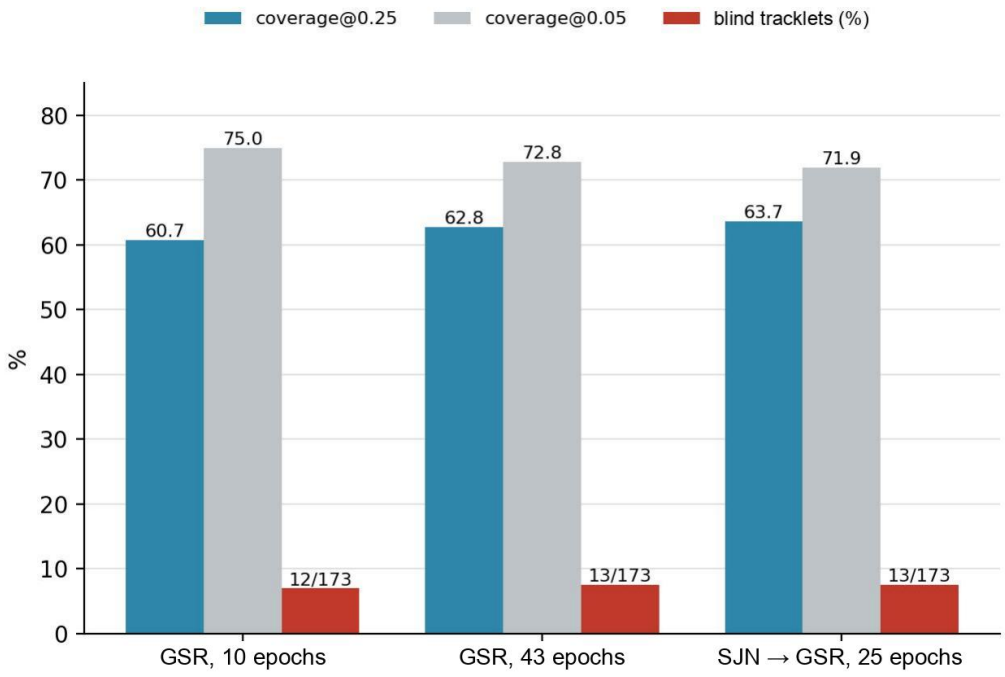


Figure 5.4: Coverage of the three jersey-number region detector checkpoints, obtained from the same set of annotated images while varying the training duration.

region is detected changes from 12 to 13 out of 173 and remains unchanged for the last checkpoint. The comparison concerns the 173 tracklets that form the common evaluation surface of the three checkpoints, for which the corresponding set of crops used in this experiment is available.

For the checkpoints considered, additional training therefore does not expand the set of cases on which the detector is able to activate. It mainly increases the confidence of regions that the model was already able to detect, moving some of them above the operational threshold without recovering the more difficult cases. Since the relevant limitation for the recognizer is precisely the absence of regions in these cases, none of the alternative checkpoints is promoted.

The comparison with the previous subsection also reveals a different trans-

fer behavior for pre-training. The external corpus produces a substantial increase in recognizer accuracy, but does not produce a comparable increase in detector coverage. The available evidence therefore suggests that transfer is effective for interpreting the digits once they have been localized, but does not solve the problem of detecting the jersey-number region in images from the target domain.

5.4.3 Crop Budget and Input Resolution

A second possibility is that the detector has too few opportunities to observe the jersey number. In the standard configuration, the crops from each tracklet are ranked according to a quality measure and the five best are retained. A minimum spacing of five frames is also enforced to limit the presence of nearly duplicate images.

To measure the effect of this constraint, both parameters are removed. The maximum number of crops becomes effectively unlimited and the minimum frame spacing is set to zero. The experiment is conducted on the three external broadcast clips used for the out-of-domain analysis, while keeping every other pipeline component unchanged.

Removing the budget increases tracklet-level coverage in all three clips: from 50.0% to 72.2% on Inter–Atalanta, from 27.4% to 56.5% on Inter–Juventus, and from 18.2% to 40.9% on USA–Belgium. A larger number of observations therefore substantially increases the probability that at least one crop in a tracklet produces a usable region.

This benefit does not correspond to a systematic improvement of the detector on individual crops. On USA–Belgium, for example, per-crop coverage decreases from 3.6% to 3.0%, while tracklet-level coverage more than doubles. The increase therefore results mainly from the greater number of available

attempts.

The results also suggest that the criterion used to select the five crops is not perfectly aligned with jersey-number visibility. On Inter–Atalanta and Inter–Juventus, per-crop coverage increases when images ranked lower by the selection criterion are also included.

The interpretation of this experiment requires three qualifications:

- **Domain.** The comparison was conducted on the three external broadcast clips rather than on SoccerNet-GSR.
- **Computational cost.** Processing all crops substantially increases the number of images passed to the module, from 77 to 944 on Inter–Atalanta, from 270 to 2849 on Inter–Juventus, and from 110 to 1447 on USA–Belgium.
- **Accuracy of new emissions.** Higher coverage does not necessarily imply higher accuracy. Among the 18 additional tracklets recovered on Inter–Juventus, a CTC confidence threshold of 0.75 almost perfectly separates the manually verified cases, with ten correct readings out of ten above the threshold. The same behavior does not emerge, however, for the five recovered tracklets considered on USA–Belgium.

The evidence is therefore insufficient either to introduce a generalizable confidence filter or to promote the processing of all crops in the end-to-end configuration, which retains the original budget.

A final experiment tests whether the small size of the detected regions constitutes a separate limitation. Before recognition, bicubic upscaling by a factor of four is applied, similarly to the preprocessing already used by the primary OCR path. On one complete Inter–Atalanta sequence, none of the

Table 5.4: Effect of removing the crop budget on the three broadcast clips. Region coverage measures the proportion of crops for which a region is detected; tracklet coverage measures the proportion of tracklets that receive at least one reading.

	Five crops per tracklet			*All crops*		
Clip	Crops	Region	Tracklet	Crops	Region	Tracklet
Inter–Atalanta	77	28.6%	50.0%	944	38.2%	72.2%
Inter–Juventus	270	11.1%	27.4%	2849	16.6%	56.5%
USA–Belgium	110	3.6%	18.2%	1447	3.0%	40.9%

thirteen tracklets with an available reading changes its predicted number, and the confidence variations are negligible.

Upscaling is therefore not adopted. When applied after region localization, it changes the representation of the available pixels but cannot recover details that are absent from the original crop.

5.4.4 Frame Selection

Increasing the number of crops shows that a tracklet may contain useful information beyond the five frames selected by the standard configuration. It remains to be determined whether part of the same benefit can be obtained without increasing the budget, by selecting more informative frames.

Since the jersey number is mainly visible when the athlete is facing away from the camera, two strategies are evaluated for estimating orientation: a classifier trained to recognize an unobstructed back view and a generic pose estimator, from which orientation is derived from the relative positions of the shoulder keypoints.

The pose estimator returns usable keypoints for 88% of the crops, 763 out

of 865, despite athlete images ranging from approximately 25 to 70 pixels in size. Orientation is strongly associated with the presence of a detectable region: the detector activates on 81.0% of crops classified as back-facing, 410 out of 506, compared with 29.0% of front-facing crops, 71 out of 245.

This relationship does not, however, translate into recovery of tracklets that contain no detected region at all. Neither strategy recovers any of the 12 tracklets for which the detector produces no region in the available frames.

Analysis of these cases reveals two distinct causes:

- **Absence of a useful view.** In 7 tracklets, none of the 20 sampled frames shows the athlete from the back. The required information is therefore absent even before the selection stage.

- **Detector failure.** In the remaining 5 tracklets, at least one frame has a favorable orientation, with shoulder-keypoint confidence values between 0.93 and 0.99, but the detector still fails to identify the jersey-number region.

The same pattern appears in a larger set of 216 crops without a detection. In 169 cases, the athlete is not facing away from the camera, while in the remaining 47 cases the orientation is favorable but the detector still fails.

Frame selection is therefore not the determining limitation. Orientation helps identify observations that are statistically more favorable, but it does not recover the tracklets responsible for complete abstention. In some cases, a useful view is absent, while in others the failure persists even when such a view is available.

Overall, the experiments on the region-based path identify two limitations:

- **Recognition.** When a region is available, external pre-training substantially improves digit recognition, increasing overall accuracy from 32.2% to 57.1%.
- **Region availability.** The detector remains the main upstream limitation. Longer training does not recover difficult tracklets, upscaling does not change the predictions, and orientation-based selection does not eliminate complete abstentions.

Increasing the crop budget is the only intervention that clearly improves tracklet coverage, but this result was observed only out of domain, involves a substantially higher computational cost, and does not provide sufficient evidence about the accuracy of the new emissions. The end-to-end configuration therefore retains the original budget.

The pre-trained recognizer thus has substantially better reading capability, but can exploit it only when the preceding stage succeeds in localizing the jersey number. Since the pipeline also includes a second independent source of numerical evidence, the primary OCR path, the following section analyzes how the two decisions should be combined when both are available.

5.5 Evidence Arbitration

Section 5.4 showed that the two recognition paths have complementary properties. The region-based recognizer achieves good accuracy when it receives a usable region, but its coverage is limited by the detector that precedes it. The primary OCR path instead provides a second source of evidence, with a different profile of errors and abstentions.

The presence of two recognizers therefore introduces a problem that is distinct from the quality of the individual models: when both produce con-

flicting outputs, it is necessary to determine which evidence should take priority. In the proposed system, each recognizer produces a record associated with its own source, and a decision layer resolves the available evidence according to a configurable ordering.

This separation makes it possible to modify the decision policy without retraining the models and while keeping every other pipeline component unchanged. The experiment compares three configurations:

- **baseline**: only the primary OCR path is used;
- **arm A**: the primary OCR path has priority, and the region-based recognizer is used only when the first source abstains;
- **arm B**: the region-based recognizer has priority, and the primary OCR path is used as a fallback.

Arm A therefore represents a conservative policy: the second recognizer can increase coverage, but it cannot replace a reading already produced by the primary OCR path. Arm B allows such replacement and can therefore both correct an incorrect reading and introduce a regression.

The policies and their corresponding metrics were defined before observing the joint results on the seven-sequence block reserved for this comparison.

5.5.1 Offline Comparison of the Policies

The first evaluation isolates arbitration from front-end errors. Bounding boxes and tracklet identities are provided by the *ground truth*, so that the observed differences can be attributed directly to the jersey-number decisions. The block contains 116 tracklets with an annotated and visible jersey number.

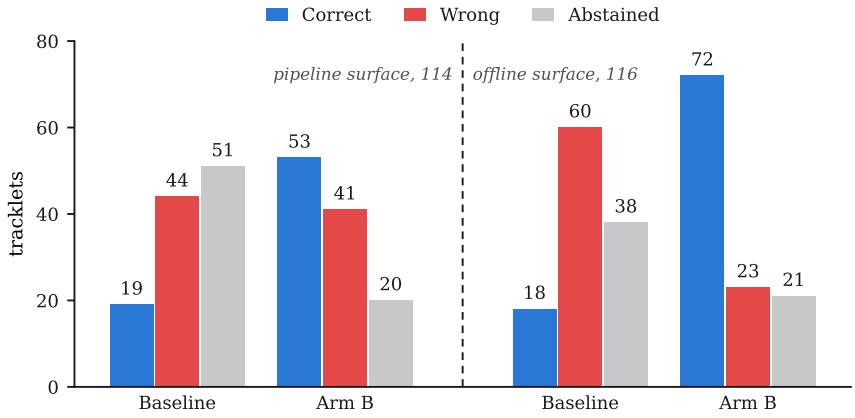


Figure 5.5: Composition of correct predictions, incorrect predictions, and abstentions for the baseline and arm B on the offline and pipeline surfaces. The offline surface uses bounding boxes and tracklet identities provided by the *ground truth*, while the pipeline surface uses the tracklets actually produced by the system. The two denominators, 116 and 114 tracklets respectively, are reported separately.

Figure 5.5 compares the baseline with arm B, while also reporting, for reference, the outcomes obtained later on the pipeline surface. arm A is not shown in the figure, but is used in the following analysis to separate the benefit due to recovering abstentions from the benefit due to replacing existing readings.

On the offline surface, the baseline produces 18 correct predictions, 60 incorrect predictions, and 38 abstentions. arm B instead reaches 72 correct predictions, with 23 errors and 21 abstentions. The overall comparison therefore shows both an increase in the number of emitted predictions and a marked reduction in incorrect readings.

arm A makes it possible to distinguish the two mechanisms. Moving from the baseline to arm A adds 10 correctly recognized tracklets by filling abstentions of the primary path. Since the region-based recognizer operates only as a fallback, it cannot modify an existing reading.

arms A and B, however, produce an output on the same 95 tracklets.

Moving from arm A to arm B increases the number of correct tracklets by 44 without changing coverage. The benefit therefore derives exclusively from the different choice of source in cases where both recognizers produce an output.

Among the 48 tracklets on which the two arms produce conflicting results, arm B is correct in 46 cases and arm A in 2. The exact binomial test gives

$$p = 8.4 \times 10^{-12},$$

while the difference in accuracy computed over the complete population of number-annotated tracklets is +0.379, with a 95% bootstrap confidence interval of

$$[+0.275, +0.496].$$

The result shows that, on this controlled surface, the main benefit does not come from using the second recognizer merely as a fallback. The conservative policy recovers some of the abstentions of the primary path, while the larger gain emerges when the region-based recognizer is allowed to replace an existing reading.

The ordering of the sources is therefore an effective component of the system. With the models, data, and inference unchanged, modifying the priority of the evidence produces a substantial change in accuracy.

5.5.2 Effect in the Complete Pipeline

The offline comparison measures the potential of the policy in the absence of upstream errors. To determine how much of this advantage remains under actual operational conditions, the same seven sequences are processed by the complete pipeline, changing only the ordering of the sources.

On this surface, the evaluation unit is an annotated tracklet that is actually reached by the end-to-end execution. The front-end covers 114 of the 116 tracklets considered in the offline evaluation, so the results from the two surfaces are kept separate.

As shown in Figure 5.5, on the pipeline surface the baseline produces 19 correct readings, 44 incorrect readings, and 51 abstentions, while arm B produces 53, 41, and 20 respectively. Accuracy on the 114 annotated tracklets therefore increases from 0.167 to 0.465, for a gain of +0.298.

The sequence-level bootstrap gives a 95% confidence interval of $[+0.208, +0.374]$. As a complementary analysis, the exact McNemar test on the discordant observations gives

$$p = 1.9 \times 10^{-8}.$$

The improvement is observed in all seven sequences in the block. The number of correct tracklets increases respectively from 6 to 11, from 0 to 1, from 2 to 9, from 4 to 9, from 4 to 9, from 2 to 9, and from 1 to 5.

The transition analysis identifies 37 recoveries: 20 abstentions converted into correct readings and 17 incorrect readings corrected by arm B. There are 3 regressions.

The benefit is therefore not explained only by higher coverage. The number of tracklets receiving a jersey number increases by 31, while the number of correct tracklets increases by 34. In addition to producing more outputs, arm B also corrects some previously incorrect readings.

Compared with the offline evaluation, however, part of the correction capability is lost. The introduction of actual detection, tracking, and linking mainly reduces how often the region-based recognizer can contribute: less favorable crops produce fewer usable regions and therefore fewer opportunities

for the higher-priority source to intervene.

arm B is therefore adopted as the final configuration. The choice is not based on the absence of regressions, but on the strong imbalance between recoveries and regressions. The conservative criterion originally considered is therefore excessively restrictive in this case, a point revisited in the conclusions.

Computational cost of arbitration. A cold-cache measurement on the same sequence shows that reversing the priority between the two sources does not appreciably change the computational cost of the pipeline. The CTC-primary configuration requires 1258.685 s in total for the two recognition stages, compared with 1265.307 s for the OCR-primary configuration, a difference of 6.622 s, corresponding to 0.52%. Both recognizers are executed in both configurations and 20 of the 24 tracklets receive an assignment. Only the priority with which the resulting evidence is used by the decision layer changes.

In the same measurement, the CTC stage requires approximately 4.8 s, while the OCR stage requires approximately 1255 s. The total runtime is therefore largely dominated by the latter. The ratio of approximately 260 to 1 between the two stages should, however, be interpreted as an observation specific to this measurement rather than as a general comparison between two isolated pipelines.

5.5.3 Roster Constraint

The source ordering determines which recognizer should take priority, but it does not yet use information that is available in advance: the list of players in the match. Since not all jersey numbers are compatible with a given team,

Table 5.5: Effect of roster-based candidate re-ranking on 36 tracklets with manually verified identities in Inter–Atalanta.

	Without roster	With roster
Correct	22	26
Accuracy	61.1%	72.2%
Transitions		
Correct in both	21	
Incorrect → correct	5	
Correct → incorrect	1	
Incorrect in both	9	

the candidates produced by the region-based recognizer can be reordered by prioritizing those present in the roster.

The evaluation requires final identity labels, which are not available for the three external broadcast clips. An execution on Inter–Atalanta was therefore manually annotated without showing the annotator model predictions, confidence scores, or evidence sources. Labels are associated with both tracklet and scene segment, so that a tracklet crossing a camera cut is not treated as a single identity.

Table 5.5 compares the assignments obtained with and without roster-based re-ranking.

Accuracy increases from 61.1% to 72.2%, with five transitions from incorrect to correct and only one in the opposite direction. The sample is limited and does not allow the general magnitude of the effect to be estimated robustly, but it shows that roster information can correct some decisions when the correct jersey number is already present among the candidates.

The behavior of the constraint can be summarized in three points:

- **benefit**: it can promote a correct candidate that has already been produced by the recognizer but is not ranked first;
- **limitation**: it cannot generate missing evidence, recover a tracklet without a detected region, or introduce a correct number that does not appear among the candidates;
- **risk**: re-ranking can replace a correct reading with an incorrect one when the prior information favors the wrong candidate.

The roster constraint therefore acts on the selection among already available hypotheses rather than on the perceptual capability of the system.

Overall, the arbitration experiments show that the benefit depends not only on recognizer quality, but also on how their evidence is used. The resulting configuration gives priority to the region-based recognizer, retains the primary OCR path as a fallback, and uses the roster to reorder compatible candidates when such information is available.

It remains to be determined how strongly this strategy depends on the domain on which the perceptual modules were developed. The following section therefore analyzes its behavior on broadcast recordings external to SoccerNet-GSR.

5.6 Out-of-Domain Evaluation

The previous experiments mainly use SoccerNet-GSR, which provides both the data used to train some components and the main evaluation surface. It therefore remains necessary to determine how strongly the observed behavior depends on the specific characteristics of that domain.

For this purpose, the pipeline is applied without modification to three external broadcast clips: Inter–Juventus, Inter–Atalanta, and USA–Belgium. The recordings differ from SoccerNet-GSR in camera geometry, athlete scale, compression, illumination, and uniform characteristics, and are processed using the same weights and configuration adopted in the previous experiments.

The clips do not provide complete positional *ground truth*. The following results therefore cannot be interpreted as an end-to-end accuracy measurement directly comparable with that obtained on SoccerNet-GSR. The analysis instead focuses on the coverage of the different recognition paths and on how the decision policy behaves when one of the sources loses the ability to produce evidence. Readings and candidates generated out of domain are not treated as correct in the absence of annotations. The results describe coverage and evidence availability, not unchanged transfer of recognition accuracy.

5.6.1 Transfer of the Region Detector

On the three external clips, the operational coverage of the region-based path is substantially lower than that observed on SoccerNet-GSR. The comparison describes the actual behavior of the module under the new recording conditions, but does not allow the entire difference to be causally attributed to domain shift in the detector alone. The external clips do not provide equivalent annotations of jersey-number visibility for individual crops.

On the frozen SoccerNet-GSR test set, the detector produces a usable region for 61.0% of the crops passed to the module. With the same budget of five crops per tracklet, the observed coverage decreases to 28.6% on Inter–Atalanta, 11.1% on Inter–Juventus, and 3.6% on USA–Belgium.

Figure 5.6 compares both per-crop coverage and the proportion of track-

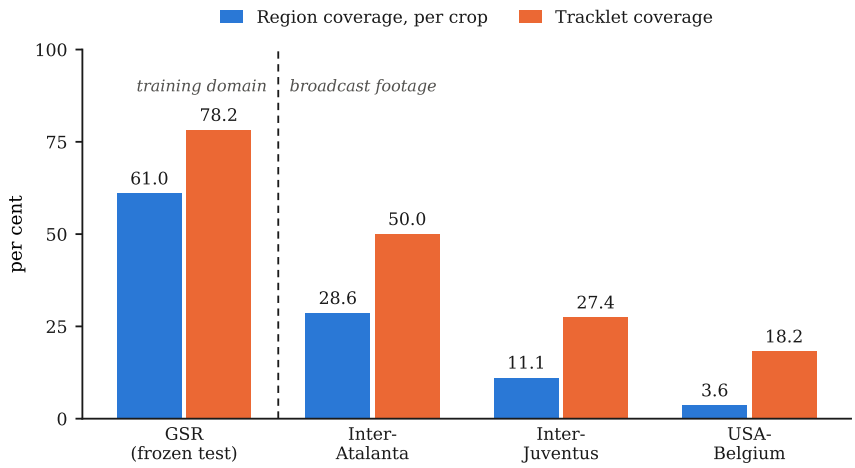


Figure 5.6: Coverage of the jersey-number region-based path within and outside the SoccerNet-GSR domain, using the same weights and the same budget of five crops per tracklet. For the external broadcast clips, the values represent coverage rather than accuracy.

lets receiving at least one reading.

Tracklet-level coverage shows the same trend. The path reaches 78.2% of number-annotated tracklets on the SoccerNet-GSR split, compared with 50.0% on Inter–Atalanta, 27.4% on Inter–Juventus, and 18.2% on USA–Belgium.

The degradation is not explained only by the crop budget. As shown in Section 5.4.3, increasing the number of observations improves tracklet-level coverage but does not eliminate the gap in per-crop detection rate relative to SoccerNet-GSR. The observed limitation therefore mainly concerns how frequently the detector is able to produce a usable region under the new recording conditions.

Overall, the region-based path therefore shows strong sensitivity to domain shift. The CTC recognizer can operate only on regions produced by the detector and, when their coverage decreases sharply, it automatically loses the opportunity to contribute to the final decision.

5.6.2 The Fallback Preserves Coverage

The primary OCR path maintains high coverage on the external clips as well: the tracklet-level assignment rate is 83.3% on Inter–Atalanta, 79.0% on Inter–Juventus, and 100% on USA–Belgium. These values describe only the availability of a reading and not its accuracy.

This property is relevant for arm B, in which the region-based recognizer has priority and the primary OCR path is used when the first source abstains. As the coverage of the CTC path decreases, a larger proportion of the available decisions therefore depends on the fallback.

The clearest case is USA–Belgium. The CTC path reaches only 18.2% of tracklets under the standard crop budget, while the primary OCR path produces a reading for 100%. The loss of coverage of the higher-priority source does not therefore translate, in this case, into a comparable loss of overall coverage.

As CTC emissions become less frequent, the behavior of arm B consequently approaches that of the OCR-based configuration. This mechanism does not necessarily preserve the accuracy gain observed on SoccerNet-GSR, but it prevents abstention by the higher-priority source from automatically causing the entire system to abstain.

When both sources produce a reading, disagreement remains frequent. On Inter–Juventus, there are 5 agreements and 10 disagreements between the two sources; 34 tracklets receive a reading from the primary OCR path only and 2 from the CTC path only. On USA–Belgium, there are 3 agreements, 1 disagreement, and 18 tracklets assigned by the primary OCR path only.

In the absence of *ground truth*, these counts do not establish which source is correct. They only describe the relative availability of the evidence and how often the policy must resolve a conflict.



1359	1467	1494	1499
38×87	55×77	48×82	56×83

Figure 5.7: Four regions used by the recognizer for a tracklet from Inter–Juventus. The frame index and the original crop dimensions (width × height, in pixels) are reported below each image.

5.6.3 Qualitative Analysis of a Conflict

A case from Inter–Juventus provides a concrete example of the type of information on which the system operates. Figure 5.7 shows the four crops from the same tracklet in which the region detector is able to localize a possible jersey number.

The jersey number is visually readable as 95. The primary OCR path returns 2, while the region-based recognizer returns 55 with confidence 0.998. Its ordered list of the five highest-scoring candidates is

$$[55, 95, 5, 25, 15].$$

The case illustrates three properties that are also observed in the quantitative experiments. The first is the scarcity of evidence. Only four crops from the entire trajectory produce a usable region, and each has a width of less than approximately sixty pixels. The problem is therefore not only to

classify a digit correctly, but also to obtain access to a sufficient number of informative observations.

The second property concerns confidence. The incorrect prediction 55 receives a value of 0.998. In this case, a policy based simply on comparing confidence scores would therefore have favored the incorrect answer. This result is consistent with the choice to keep arbitration based on the source and on explicit rules, rather than directly treating scores produced by different models as quantities calibrated with respect to one another.

The third property is that the correct answer is not absent from the recognizer: 95 appears in second position among the candidates. This is the type of situation in which the roster constraint analyzed in Section 5.5.3 can intervene, promoting a candidate compatible with the match roster without requiring the pixels to be processed again.

The qualitative case does not constitute an accuracy measurement, but it shows how the different mechanisms analyzed in the previous sections interact on the same tracklet: limited coverage, confidence that is not necessarily calibrated with correctness, conflicting sources, and the use of external knowledge to select among candidates that are already available.

5.6.4 Implications of Domain Shift

The out-of-domain analysis mainly reveals a loss of coverage in the region detector, which decreases from 61.0% per crop on SoccerNet-GSR to values between 28.6% and 3.6% on the three external clips. The results can be summarized in three points:

- **Observed limitation.** The reduced availability of regions directly limits the contribution of the CTC path. In the absence of complete

ground truth, it is not possible to determine whether the recognizer's accuracy, when a region is available, transfers unchanged.

- **Front-end interventions.** Increasing the number of crops improves tracklet coverage but does not eliminate the per-crop gap. Neither upscaling nor frame selection solves the missed-detection cases. The results therefore indicate adapting the detector to a wider variety of broadcast productions as the priority intervention.

- **Role of the fallback.** When the CTC path abstains, the decision can fall back to the primary OCR path. The gain observed on SoccerNet-GSR is therefore not necessarily preserved out of domain, but failure of the higher-priority source does not automatically cause the system to abstain.

It remains to be determined whether the selected configuration produces an improvement on the complete task. The following section performs this comparison using GS-HOTA.

5.7 End-to-End Evaluation

The previous sections analyzed the main stages of the pipeline separately and led to the selection of an operational configuration: TVCalib for field localization, the pre-trained CTC recognizer, the region-based path as the higher-priority source, and the primary OCR path as a fallback.

It remains to be determined whether the improvement observed in jersey-number arbitration translates into a benefit on the complete task. The evaluation is performed using GS-HOTA, the metric adopted by the SoccerNet-GSR benchmark to jointly measure localization, temporal association, and

identity attributes.

5.7.1 Interpretive Reminder on GS-HOTA

GS-HOTA is formally defined in Chapter 4. For the purpose of interpreting the following results, it is sufficient to recall that it combines localization, temporal association, and correctness of the identity attributes. The multiplicative nature of the metric implies that an error in one of the required attributes can make a prediction ineligible for matching even when position and tracking are correct. An improvement in jersey-number recognition can therefore have an effect larger than the change in its standalone accuracy.

The spatial component is also relatively strict: the mean error of 1.43 m obtained by TVCalib in Section 5.2 already produces a non-negligible reduction in similarity. For this reason, all GS-HOTA evaluations reported below use TVCalib rather than the fallback based on the field quadrilateral.

5.7.2 Results on the Validation Pilot

The end-to-end evaluation compares arms A and B on the twelve-sequence validation pilot, which is distinct from the seven sequences used to analyze the decision policy directly in Section 5.5. The only difference between the two configurations is the priority assigned to the jersey-number sources; all other stages of the pipeline remain unchanged.

Figure 5.8 reports GS-HOTA and its main components.

GS-HOTA increases from 0.2097 for arm A to 0.2870 for arm B, with a difference of

$$\Delta\text{GS-HOTA} = +0.0772,$$

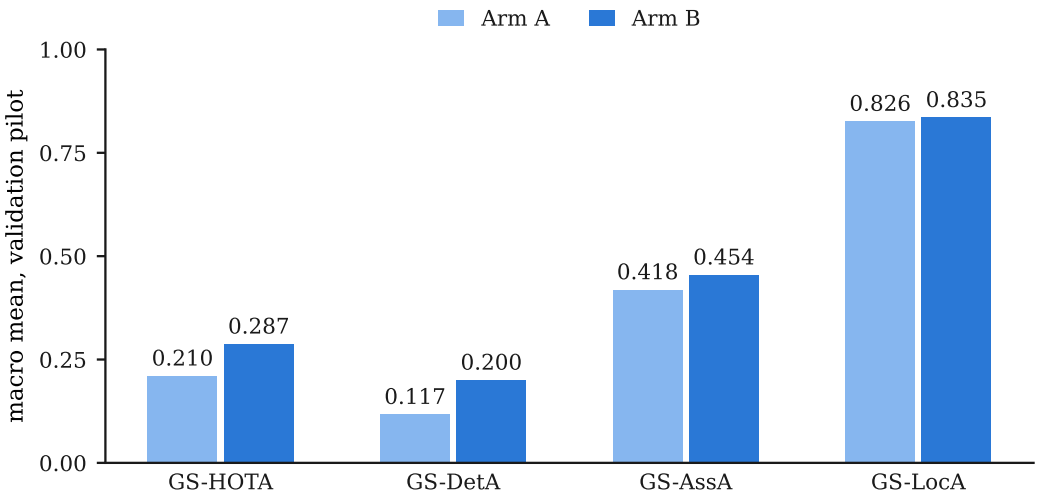


Figure 5.8: End-to-end comparison between arm A and arm B on the twelve-sequence validation pilot. Differences are paired by sequence and the 95% confidence intervals are obtained through sequence-level bootstrap.

and a 95% confidence interval of

$$[+0.0476, +0.1067].$$

The benefit of arbitration is therefore also observable on the end-to-end task.

The decomposition of the metric shows increases in all components:

- **GS-DetA**: +0.0829, with interval $[+0.0419, +0.1293]$;
- **GS-AssA**: +0.0358, with interval $[+0.0174, +0.0557]$;
- **GS-LocA**: +0.0092, with interval $[+0.0017, +0.0190]$.

These changes do not indicate an improvement in detection, tracking, or calibration, which are identical in the two arms. They instead result from the GS-HOTA matching procedure: correcting the jersey number makes already

correctly detected and tracked tracklets eligible for matching, producing in particular the observed increase in GS-DetA.

The change in GS-LocA is also compositional. The coordinates are identical, but the set of valid matched pairs over which the component is computed changes. It therefore does not represent an improvement in calibration.

5.7.3 Comparison with the Published Baseline

As an external reference, the baseline published with SoccerNet-GSR reports a GS-HOTA value of 0.1805 on the validation split. On the twelve-sequence pilot used in this work, arm A reaches 0.2097 and arm B reaches 0.2870.

These values provide a reference for the order of magnitude of the performance, but they do not constitute a controlled comparison because the experimental conditions differ in several respects:

- the results in this thesis are obtained on a subset of twelve sequences rather than on the complete validation partition;
- GS-HOTA is computed using an independent implementation of the published definition;
- some components of the proposed pipeline are fine-tuned on the target domain, whereas the baseline was designed as a more general reference.

The published value is therefore used only as an external reference. The main experimental evidence remains the paired comparison between arm A and arm B, in which only the priority of the jersey-number sources changes.

5.7.4 Confirmation on the Test Split

The selection of arm B is finally verified on a separate block of twelve sequences from the SoccerNet-GSR test split.

The test split had already been used in the offline recognizer experiment described in Section 5.4.1. The present experiment serves a different purpose: to verify end to end the arbitration policy selected on the validation pilot.

The protocol for the final verification was fixed before execution:

- twelve of the 49 sequences were selected using a recorded seed;
- arms A and B and the corresponding arbitration policy had already been defined on validation;
- the results on this block were not used to modify the source ordering or introduce further changes to the end-to-end configuration.

The selection of arm B therefore could not be changed based on the result observed on this block. In this comparison as well, the two configurations differ only in the priority assigned to the jersey-number sources; detection, tracking, calibration, team, and role remain unchanged.

The macro-average GS-HOTA increases from 0.2759 to 0.3235, corresponding to a gain of

$$\Delta\text{GS-HOTA} = +0.0476.$$

Sequence-level bootstrap gives a 95% confidence interval of

$$[+0.0262, +0.0685],$$

which excludes zero.

The improvement is also distributed across sequences:

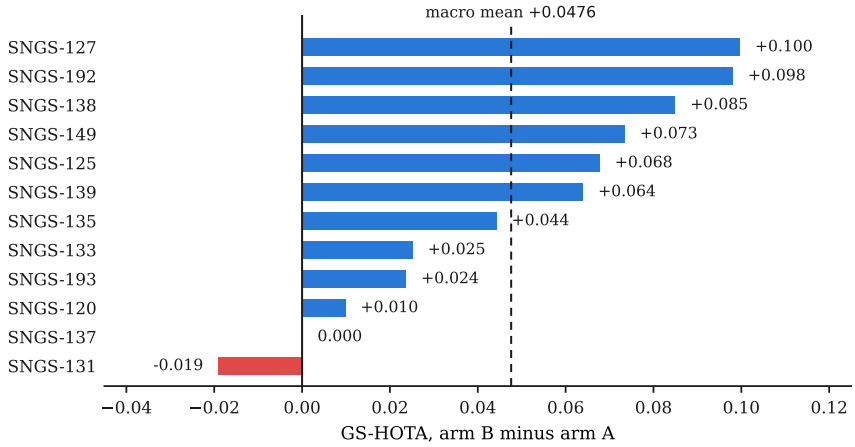


Figure 5.9: Difference in GS-HOTA between arm B and arm A on the twelve test sequences, ordered by magnitude. Ten sequences favor arm B, one is tied, and one favors arm A.

- ten sequences favor arm B;
- one sequence produces an exact tie;
- one sequence favors arm A.

The two-sided sign test on the eleven non-tied sequences gives

$$p = 0.012.$$

Figure 5.9 shows the distribution of the per-sequence differences.

The decomposition of the metric is also consistent with the behavior observed on validation:

- **GS-DetA**: +0.0566, with interval $[+0.0293, +0.0838]$;
- **GS-AssA and GS-LocA**: the confidence intervals include zero.

Since the two configurations share association and calibration, this behavior is consistent with the interpretation provided for the validation pilot.

The effect on the test block, +0.0476, is smaller than the +0.0772 observed on validation, but the corresponding confidence intervals overlap. The two blocks therefore do not provide contradictory results, although the effect size is reduced on the final test block.

Two sequences require specific interpretation:

- **SNGS-125.** A total of 56.5% of the annotations lack field coordinates and therefore cannot contribute to GS-HOTA. The sequence is retained because it was included in the protocol before observing the results. If it were excluded a posteriori, the aggregate difference would be +0.0458, which is close to the reported value.

- **SNGS-137.** The tie does not correspond to identical jersey-number performance. The team classifier produces no assignments for the players, making them ineligible for matching regardless of jersey-number quality. The residual score of 0.0925 comes almost entirely from referees. On emitted readings, arm B instead reaches 60.9% accuracy, compared with 46.4% for arm A.

SNGS-137 therefore provides a concrete example of the multiplicative nature of GS-HOTA: an improvement in one attribute may not be reflected in the metric when another attribute required for matching is missing.

5.7.5 Verification of Attribution on the Test Block

The ablation from Section 5.3 is repeated on the twelve test sequences to verify the stability of the diagnosis obtained on the validation pilot.

The main result is reproduced:

- with the actual jersey number, the difference between arm B and arm A ranges from 0.052 to 0.084 GS-HOTA points;

Table 5.6: Isolated cost of the main modules in GS-HOTA points for arm B, comparing the validation pilot and the test block.

Module	Validation	Test
Jersey number	0.281	0.226
Detection and tracking	0.257	0.246
Calibration	0.206	0.231
Team	0.181	0.081
Role	0.161	0.029

- when the jersey number is replaced with the *ground truth*, the gap decreases to values between 0.001 and 0.002;
- with all components set to oracle, GS-HOTA reaches exactly 1.000 for both arms.

The near disappearance of the gap when the jersey number is oracle therefore confirms on the test block as well that the difference between the configurations is attributable to the handling of numerical evidence.

The isolated costs of the individual modules are reported in Table 5.6.

The comparison between the two blocks reveals two behaviors:

- **stability**: jersey number, detection and tracking, and calibration retain costs of the same order of magnitude;
- **variability**: the costs associated with team and role decrease substantially on the test block.

The second difference is mainly associated with abstentions. On the validation pilot, the team classifier leaves the attribute empty for approximately half of the players in six of the twelve sequences, whereas on the test block

this behavior occurs in only one sequence. When the module produces an output, accuracy on the emitted labels is generally above 90%.

Attribution therefore also depends on the tendency of the modules to abstain. The conclusion that remains stable across both blocks is that the jersey number, detection and tracking, and calibration contribute on a comparable scale to the gap from ideal performance, with no single dominant bottleneck.

5.7.6 Evaluation Conclusion

The end-to-end evaluation confirms the advantage of the selected policy:

- on the validation pilot, arm B increases GS-HOTA by +0.0772;
- on the final test block, the gain remains positive at +0.0476, with ten favorable sequences, one tie, and one unfavorable sequence;
- the test ablation confirms that the effect is attributable to the handling of the jersey number, since the gap between the arms becomes negligible when this attribute is replaced with the *ground truth*.

The result is obtained by changing only the priority of the evidence, without modifying the models or the shared stages of the pipeline. The final test block was not used to select or revise the arbitration policy chosen on validation; arm B therefore remains the promoted configuration of the pipeline.

Chapter 6

Conclusions

This thesis addressed two complementary problems. The first is practical and concerns player identification from broadcast footage by combining heterogeneous evidence that is not always available. The second is methodological and concerns identifying the stages that actually limit the performance of a pipeline composed of multiple interacting modules.

The results show that identification does not depend exclusively on the quality of the individual models. The way evidence is produced, made available, and arbitrated can have a measurable impact on the final outcome. At the same time, oracle substitutions show that the gap from ideal performance cannot be attributed to a single bottleneck: jersey-number recognition, detection and tracking, calibration, and the availability of other attributes contribute to the overall result in different ways.

6.1 What the Experiments Established

The experiments produced five main findings:

- **Evidence arbitration.** Changing only the priority between the two

recognizers produces a consistent end-to-end improvement on both validation and test data. On the final test block, arm B improves GS-HOTA by +0.0476 and outperforms arm A on ten of the twelve sequences. The transition analysis also shows a clear imbalance in favor of recoveries over regressions, indicating that the benefit is not explained only by increased coverage.

- **Recognizer pre-training.** External pre-training followed by fine-tuning on SoccerNet-GSR substantially improves jersey-number recognition, increasing overall accuracy from 32.2% to 57.1%. Since the compared checkpoints operate on the same detected regions, the improvement concerns recognition quality rather than evidence availability.

- **Roster constraint.** The manually annotated experiment shows that roster information can correct some identity assignments when the correct jersey number is already present among the recognizer hypotheses. Its effect is promising, although the limited sample size does not support a precise estimate of its general impact.

- **Absence of a single bottleneck.** Oracle substitutions show that jersey-number recognition, detection and tracking, and calibration contribute on a comparable scale to the remaining error. The gap from ideal performance therefore cannot be attributed to a single clearly dominant component.

- **Limitation in tracklet association.** Detection quality is substantially stronger than temporal association, and tracklet linking does not improve tracking performance under the evaluated configuration. The experiments further show that more accurate geometry alone cannot

solve the problem, because in many failure cases the correct merge is not generated among the available candidates.

Overall, the most relevant result is that part of the performance can be recovered without modifying the models, but by changing how the evidence they produce is used. The controlled comparison between arm A and arm B provides the main evidence for this effect. The comparison with the published SoccerNet-GSR baseline is instead retained only as a contextual reference, since the evaluation surface, pipeline components, and metric implementation differ.

6.2 A Recurring Pattern Across the Results

The experiments reveal a recurring distinction between the quality of the evidence and the way it is made available to and used by the final decision. This pattern emerges in three main forms:

- **Evidence quality and use.** Pre-training improves the quality of the recognizer predictions, arbitration changes the priority between two already available sources, and the roster constraint introduces knowledge about the identity space. In these cases, the improvement does not require changes to the detection and tracking front-end.

- **Upstream loss of evidence.** More accurate calibration cannot improve linking when the correct merge is not generated among the candidates. Similarly, an accurate recognizer cannot contribute when the detector fails to localize the region containing the jersey number. Improving an isolated module therefore does not guarantee an end-to-end benefit if the required information is lost before reaching it.

- **Out-of-domain transfer.** Per-crop coverage of the region detector decreases from 61.0% on SoccerNet-GSR to 28.6%, 11.1%, and 3.6% on the three external clips, while the alternative OCR path continues to produce evidence for a substantial proportion of the tracklets. In the absence of complete annotations, out-of-domain recognizer accuracy cannot be measured, but region availability emerges as the main observable operational limitation of the region-based path.

Pipeline design therefore cannot focus only on the accuracy of individual modules. It is also necessary to consider which evidence reaches the decision layer and how that evidence is used. The separation between observation, evidence, and decision is thus the architectural principle that made it possible both to improve the system and to experimentally attribute the causes of the observed improvements.

A further consequence concerns the criterion used to accept a new configuration. The initial requirement of introducing no regressions proved to be excessively restrictive: moving from arm A to arm B produces 3 regressions but 37 recoveries, while in the pipeline the number of correct predictions increases from 19 to 53 and the number of incorrect predictions decreases from 44 to 41. In the presence of abstention, recoveries and regressions must therefore be considered jointly, while preserving the option to leave an identity unassigned when the evidence is insufficient.

6.3 Limitations

The results should be interpreted within the limits of the experimental protocol used in this work.

Sample size. The most computationally expensive evaluations are conducted on subsets of the benchmark. The final end-to-end comparison uses twelve test sequences, while some manual analyses, such as the evaluation of the roster constraint, involve even smaller samples. Confidence intervals and paired tests quantify part of the uncertainty, but they do not replace an evaluation on the complete benchmark.

Use of the test split. The test split is used at two distinct stages. The first concerns the offline comparison between recognizer checkpoints; the second verifies end to end, on twelve sequences sampled according to a frozen protocol, the decision policy selected on validation. The second use is not employed to select the source ordering, but this distinction must be preserved when interpreting the results.

End-to-end metric. GS-HOTA combines multiple attributes and strongly penalizes the absence of any required one. A local improvement may therefore not be visible in the final score when another component still prevents correct matching. Sequence SNGS-137 provides an example: arm B improves jersey-number recognition, but the absence of team predictions prevents the improvement from being reflected in GS-HOTA.

Oracle-based attribution. Oracle substitutions are applied after inference. The isolated costs therefore measure how strongly the error of a component affects the observed output, rather than how the entire pipeline would behave if that component were actually perfect during execution. In addition, the costs of the different stages are not additive because their interactions influence the final result.

Out-of-domain evaluation. The three additional broadcast clips do not provide complete *ground truth* equivalent to SoccerNet-GSR. They make it possible to measure coverage, evidence availability, and agreement between sources, but not to estimate general out-of-domain identification accuracy. Only the manually annotated Inter–Atalanta subset allows identity to be evaluated directly.

Interpretation of the mechanisms. Some explanations proposed by the experiments are interpretations supported by the data rather than causally isolated effects. In particular, the loss of coverage of the region detector may jointly depend on scale, uniforms, typefaces, compression, and recording characteristics. Similarly, failure of tracklet linking does not allow the error to be attributed to a single internal mechanism, even though candidate analysis shows that some correct merges are excluded before the ranking stage.

6.4 Future Work

The results identify three priority directions for future development.

Diagnosing abstention in team classification. The test evaluation shows that the classifier can be accurate when it emits a prediction, but has a high abstention rate on some sequences. Since missing team information can prevent end-to-end matching even when the other attributes are correct, identifying the cause of these abstentions represents a relatively focused intervention with potential impact on the entire pipeline.

Reducing dependence on the specialized region detector. The main operational limitation observed out of domain for the CTC path is the loss

of coverage of the jersey-number region detector. One possible direction is to increase the diversity of the training data by including different broadcast styles, uniforms, and scales. A second is to reduce dependence on a specialized detector, for example by using geometric information derived from pose estimation to directly identify a candidate region. Both approaches should be evaluated explicitly out of domain.

Improving candidate generation for tracklet linking. The calibration experiments show that more accurate geometry provides no benefit when the correct merge does not reach the ranking stage. A natural direction is therefore to expand or redesign candidate generation and then use appearance and geometry to rank the resulting candidates. Such an intervention would act directly on the association component, which remains one of the main sources of front-end error.

6.5 Final Summary

The contribution of this thesis lies in the construction and analysis of a pipeline in which observation, evidence, and decision remain explicitly separated. This separation made it possible not only to improve the system, but also to trace the results more precisely to the decisions and components that produced them.

The experimental evaluation shows that the gap from ideal performance arises from the interaction of multiple stages of the pipeline. At the same time, the comparison between arms A and B demonstrates that part of the performance can be recovered without modifying the models, by changing how already available evidence is used in the final decision.

Player identification from broadcast video therefore emerges as a problem

of integrating incomplete, heterogeneous, and sometimes conflicting evidence, rather than as a simple sequence of independent classifiers. Making the transition from evidence to decision explicit makes it possible both to obtain better identifications and to understand more precisely where and why the pipeline fails.

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